

Part III

Techniques for Removal of Surface Contamination

Chapter 11

Cleaning with Solvents

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11.1 INTRODUCTION

Cleaning with solvents dates back to ancient times, possibly beginning with the advent of textile clothing. The ruins of Pompeii reveal a highly developed trade of “fullers,” professional clothes cleaners who used “fuller’s earth,” a type of clay, to absorb soils and grease from clothing too delicate for laundering with lye or ammonia.

There are many stories about the origin of dry cleaning, all based on a surprise discovery when a petroleum-type fluid was accidentally spilled on a greasy fabric. The fluid quickly evaporated and the stains were miraculously removed. This was solvent cleaning in its simplest form.

The ability to clean with solvents is valued quite highly by some. In August 2003, a citizen of Florida pleaded guilty to violating the U.S. Clean Air Act when he attempted to smuggle ninety 30-pound cylinders of CFC-113 and CFC-12. He had intended to sell the material to aircraft maintenance facilities

as a cleaning solvent and to local automotive repair shops as a replacement refrigerant for vehicular air conditioning systems. The street value of his shipment was nearly U.S. \$70,000. That is U.S. \$26/pound!

This chapter is focused on the solvent-cleaning *process*, rather than on just the solvent. The role of well-chosen solvents is to fulfill the requirements of that process. It is the *process* (washing, rinsing, and drying) which provides effective cleaning. The solvent plays a vital role; the process would not function without it. But it is the overall solvent-cleaning process, not just the solvent, which must command attention.

The material in this chapter is neither solely practical nor solely theoretical. It is not an exhaustive or encyclopedic treatise on solvent cleaning, but a brief overview of a broad topic that provides resources for those with further interest. A full discussion of this practical and theoretical approach can be found in this author's recent book.¹

11.2 ENVIRONMENTAL AND REGULATORY ISSUES

The capability to clean with solvents has become limited by national and international laws. In the U.S., the National Environmental Policy Act (NEPA) of 1969; 42 U.S.C. 4321-4347 is the basic national charter for protection of the environment. It establishes policy, sets goals, and provides means for carrying out the policy.

More than 15 distinct major laws (not regulations) can and do impact those doing solvent cleaning work, which often raise more concern than does resolution of problems founded in technology. Included are:

1. Chemical Safety Information, Site Security and Fuels Regulatory Relief Act, Public Law 106-40, Jan. 6, 1999; 42 U.S.C. 7412(r).
2. The Clean Air Act (CAA); 42 U.S.C. s/s 7401 et seq. (1970).
3. The Clean Water Act (CWA); 33 U.S.C. ss/1251 et seq. (1977).
4. Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA or Superfund); 42 U.S.C. s/s 9601 et seq. (1980).
5. The Emergency Planning & Community Right-To-Know Act (EPCRA); 42 U.S.C. 11011 et seq. (1986).
6. The Endangered Species Act (ESA); 7 U.S.C. 136; 16 U.S. C. 460 et seq. (1973).
7. Federal Insecticide, Fungicide and Rodenticide Act (FIFRA); 7 U.S.C. s/s 135 et seq. (1972).
8. Federal Food, Drug, and Cosmetic Act (FFDCA); 21 U.S. C. 301 et seq. (1934).
9. Food Quality Protection Act (FQPA); Public Law 104-170, Aug. 3, 1996.
10. The Freedom of Information Act (FOIA); U.S.C. s/s 552 (1966).
11. The Occupational Safety and Health Act (OSHA); 29 U.S. C. 651 et seq. (1970).
12. The Oil Pollution Act of 1990 (OPA); 33 U.S.C. 2702 to 2761.

13. The Pollution Prevention Act (PPA); 42 U.S.C. 13101 and 13102, s/s et seq. (1990).
14. The Resource Conservation and Recovery Act (RCRA); 42 U.S.C. s/s 321 et seq. (1976).
15. The Safe Drinking Water Act (SDWA); 42 U.S.C. s/s 300f et seq. (1974).
16. The Superfund Amendments and Reauthorization Act (SARA); 42 U.S.C. 9601 et seq. (1986).
17. The Toxic Substances Control Act (TSCA); 15 U.S.C. s/s 2601 et seq. (1976).

The use of solvents in cleaning and other applications encroaches on the land, sea, and air, and on the bodies of humans who use the solvents or otherwise are exposed to them. But unlike the formulation and application of paints and coatings or completion of chemical reactions in solvents or carrier fluids, solvent cleaning may consume only a small volume of solvent product. Still, only seldom have regulators exempted an application because its use volume was minuscule (*de minimus*). One exception to this policy is that the U.S. South Coast Air Quality Management District, in Rule 1171—Solvent Cleaning Operations, Section (h) (3) (F), exempted medical device manufacturers and pharmaceutical facilities using (emitting) up to 1.5 gallons per day of VOC (volatile organic compound) solvents.²

11.2.1 Ozone-Depleting Solvents (Chemicals)

Some compounds (solvents) are so inert that they survive in the lower atmosphere and populate the earth's upper atmosphere. Various zones are identified in the earth's atmosphere by altitude:

- Troposphere (surface to 10 km)
- Stratosphere (10–40 km)
- Mesosphere (40–50 km)
- Thermosphere (50–300 km)
- Exosphere (300–400 km)

Solvents, whether used for cleaning or other purposes, have significant effects within the troposphere and stratosphere.

A generation ago, scientific data showed that chlorine atoms in these compounds could be liberated by reaction with high-intensity UV (ultraviolet) light from the sun. In the stratosphere, these chlorine atoms react with ozone and consume it. These chemicals, containing chlorine (or bromine) atoms, are called ozone-depleting chemicals (ODCs).

It is the chlorine that makes a substance ozone-depleting.

- CFCs (chlorofluorocarbons, compounds containing chlorine, fluorine, carbon, and possibly hydrogen) and HCFCs (hydro chlorofluorocarbons, compounds containing chlorine, fluorine, carbon, and hydrogen) are a threat to the ozone layer.

- HFCs (hydrofluorocarbons, compounds containing fluorine, carbon, hydrogen, and no chlorine) and hydrofluoroethers (HFEs) are not a threat to the ozone layer.
- CFC-113 is a strong ODC not because it contains three fluorine atoms, but because it contains three chlorine atoms.
- Carbon tetrachloride has a very strong ODP (ozone-depletion potential of 1.1) because it contains four chlorine atoms.

Fortunately, most molecules with chlorine atoms are fairly reactive. They degrade within 6–8 days (trichloroethylene) and 5–6 months (perchloroethylene and methylene chloride).³ They are regarded as low tropospheric ozone creators as well as insignificant (<0.5%) contributors to acid rain formation. Their ODP is negligible and they are not regulated as ozone-depleters.

The ban on manufacture of CFC-113 was necessary, consequential, and highly ironic. After all, CFC-113 was developed to be inert. Its inertness was valued, relative to trichloroethylene (TCE), because it had negligible effect on human health. But its inertness allowed emitted CFC-113 to reach the stratosphere and react to destroy ozone.

In summary, two factors are necessary for a solvent to be of concern relative to its reactivity with the earth's ozone layer:

- a solvent must be adequately inert so it reaches the stratosphere, where
- chlorine (or bromine) atoms in the solvent molecule react with ozone.

11.2.1.1 Regulation of Ozone-Depleting Chemicals

The discovery of the annual depletion of ozone above the Antarctic was first announced in a paper by Farman et al.⁴ The British Antarctic Survey (BAS) is an excellent reference about that publication, the science behind it, and the events which followed it.⁵

Later, National Aeronautics and Space Administration (NASA) scientists re-analyzed their satellite data and found that the whole of the Antarctic was affected. NASA produced a color-enhanced photograph (Figure 11.1) which clearly identified the “ozone hole”.⁶

In 1977, the Coordinating Committee on the Ozone Layer was established by the United Nations Environment Programme (UNEP). UNEP's Governing Council adopted the World Plan of Action on the Ozone Layer. In 1981, UNEP acted on a proposal submitted by a meeting of legal experts, chaired by Canada, and decided to develop a global convention. Adding to the difficulty of achieving an agreement between the participants were questions about the validity of the atmospheric science and data. Many felt there was not the right technological capacity to respond to the challenge.⁷

In 1985, the Vienna Convention on the Protection of the Ozone Layer was signed. By September 1987, the disagreements and lack of understanding had given way to limited trust. Ultimately, however, the trust expanded and offered the prospect of consensus on control measures.

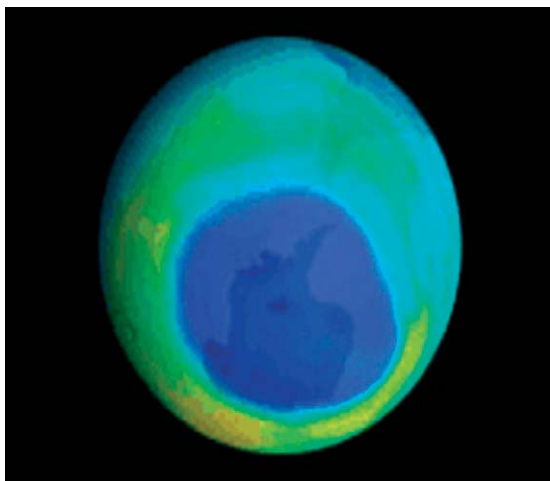


FIGURE 11.1 The “Ozone Hole.”

Thus, it was on September 16, 1987 that The Montreal Protocol on Substances That Deplete the Ozone Layer was signed by 24 countries.⁸ As of this writing, over 160 countries have signed it.⁹ The Montreal Protocol, administered by UNEP, is the international agreement to preserve the stratospheric ozone layer that protects the earth from harmful radiation. See <http://www.teap.org/> for information about the UNEP Technology and Economic Assessment Panel, which provides technical guidance about compliance to countries.

There were at least four unique characteristics of this agreement.¹⁰

- Accepted that the environmentally undesirable CFC chemicals had real value in the industrial economies of all countries as refrigerants and solvents.
- Recognized historical and economic differences among countries (developed and developing countries).
- Recognized that a ban on local use would be unfair, difficult to enforce, ultimately unproductive, and thus chose the more productive step of banning manufacture.
- Chose to emphasize some CFC chemicals versus others by both reactivity with sunlight and their emitted volume.

The ban on manufacture was an inspired regulatory choice. With one phrase the number of firms which had to be “policed” was reduced from tens of thousands to a dozen or so. The use (versus manufacture) of only one cleaning solvent, HCFC 141b, was banned by the U.S. EPA (Environmental Protection Agency).

In general, the Montreal Protocol required actions by the signing parties. Actions regarding types of ODC are shown in Table 11.1.¹⁰ Not all types of ODCs involve normally used cleaning solvents.

TABLE 11.1 Summary of Montreal Protocol Control Measures

Ozone Depleting Substances	Class	Developed Countries	Developing Countries
Chlorofluorocarbons (CFCs)	I	Phased out end of 1995 (Note 1: excluding a few uses considered essential to health or for security and/or laboratory and analytical procedures)	Total phase-out by 2010
Halons	I	Phased out end of 1993 ⁷	Total phase-out by 2010
Carbon tetrachloride	I	Phased out end of 1995 (see Note 1, above)	Total phase-out by 2010
Methyl chloroform	I	Phased out end of 1995 (see Note 1, above)	Total phase-out by 2015
Hydrobromofluorocarbons (HBFCs)	I	Phased out end of 1995	Phased out end of 1995
Methyl bromide	I	Frozen in 1995 at 1991 base level (reviewed in 2003 to decide on interim further reductions) 25% reduction by 1999	Frozen in 2002 at an average of the 1995–1998 base level
		50% reduction by 2001	20% reduction by 2005 (reviewed in 2003 to decide on interim further reductions beyond 2005)
		70% reduction by 2000	Total phase-out by 2015
			Total phase-out by 2005
Hydrochlorofluorocarbons (HCFCs)	II	Frozen from beginning of 1996 (based on 1989 HCFC consumption with an extra allowance (ODP weighted) equal to 2.8% of 1989 CFC consumption)	

Continued

TABLE 11.1 Summary of Montreal Protocol Control Measures—Cont’d			
Ozone Depleting Substances	Class	Developed Countries	Developing Countries
		35% reduction by 2004	
		65% reduction by 2010	Freeze in 2016
		90% reduction by 2015 (except for a few uses considered essential to health and/or laboratory and analytical procedures)	At 2015 base level
		Total phase-out by 2020 (up to 0.5% of base level consumption can be used until 2030 for servicing existing equipment)	Total phase-out by 2040

All reductions in manufacture include an exemption for pre-shipment and quarantine uses. Emphasis among ODCs by types was done via creation of classes and groups. Distinction among classes was made based on stratospheric reactivity. Distinction among groups, within classes, was made based on structure.

Class I substances listed in Section 602 of the 1990 Clean Air Act (CAA) include CFCs, halons, carbon tetrachloride, and methyl chloroform. The U.S. EPA later added HBFCs and methyl bromide to the list by regulation (Table 11.2).

- *Class I substance* (Table 11.2): one of several groups of chemicals with an ODP of 0.2 or higher.
- *Class II substance* (Table 11.3): a chemical with an ODP of less than 0.2. Currently, all of the HCFCs are Class II substances.

The source of ODP and GWP (global warming potential) data used in this chapter (Tables 11.2 and 11.3) is also the source used by the U.S. EPA.¹¹

Table 11.4 shows unequal treatment by the CAA for HCFC 141b and 1,1,1 TCA (trichloroethane). Yet their relative ODP values are similar: 1,1,1 TCA is a Class I ODC, yet its ODP is less than the criteria value of 0.2. The reason for this assignment is that in the late 1980s, production of 1,1,1 TCA was far

TABLE 11.2 Class I Ozone-Depleting Substances

Product Name	Chemical Name	Lifetime (years)	ODP	GWP	CAS Number
Group I					
CFC-11 (CCl ₃ F)	Trichlorofluoromethane	45	1.00	4680	75-69-4
CFC-12 (CCl ₂ F ₂)	Dichlorodifluoromethane	100	1.00	10,720	75-71-8
CFC-113 (C ₂ F ₃ Cl ₃)	1,1,2-Trichlorotrifluoroethane	85	1.00	6030	76-13-1
CFC-114 (C ₂ F ₄ Cl ₂)	Dichlorotetrafluoroethane	300	0.94	9880	76-14-2
CFC-115 (C ₂ F ₅ Cl)	Monochloropentafluoroethane	1700	0.44	7250	76-15-3
Group II					
Halon 1211 (CF ₂ ClBr)	Bromochlorodifluoromethane	16	6.00	1860	353-59-3
Halon 1301 (CF ₃ Br)	Bromotrifluoromethane	65	2.00	7030	75-63-8
Halon 2402 (C ₂ F ₄ Br ₂)	Dibromotetrafluoroethane	20	<8.6	1620	124-73-2
Group III					
CFC-13 (CF ₃ Cl)	Chlorotrifluoromethane	640	1.00	14,190	75-72-9
CFC-111 (C ₂ FCl ₅)	Pentachlorofluoroethane		1.00		354-56-3
CFC-112 (C ₂ F ₂ Cl ₄)	Tetrachlorodifluoroethane		1.00		76-12-0
CFC-211 (C ₂ FCl ₇)	Heptachlorofluoropropane		1.00		422-78-6
CFC-212 (C ₃ F ₂ Cl ₆)	Hexachlorodifluoropropane		1.00		3182-26-1
CFC-213 (C ₃ F ₃ Cl ₅)	Pentachlorotrifluoropropane		1.00		2354-6-5

Continued

TABLE 11.2 Class I Ozone-Depleting Substances—Cont'd

Product Name	Chemical Name	Lifetime (years)	ODP	GWP	CAS Number
CFC-214 (C ₃ F ₄ Cl ₄)	Tetrachlorotetrafluoropropane		1.00		29,255-31-0
CFC-215 (C ₃ F ₅ Cl ₃)	Trichloropentafluoropropane		1.00		4259-43-2
CFC-216 (C ₃ F ₆ Cl ₂)	Dichlorohexafluoropropane		1.00		661-97-2
CFC-217 (C ₃ F ₇ Cl)	Chloroheptafluoropropane		1.00		422-86-6
Group IV					
CCl ₄	Carbon tetrachloride	26	0.73	1380	56-23-5
Group V					
Methyl chloroform (C ₂ H ₃ Cl ₃)	1,1,1-Trichloroethane	5	0.12	144	71-55-6
Group VI					
Methyl bromide (CH ₃ Br)		0.7	0.38	5	74-83-9
Group VII					
CH ₂ Br ₂	Dibromofluoromethane		1.00		
HBFC-12B1 (CHF ₂ Br)	Ibromodifluoromethane		0.74		
CH ₂ FBr	Fluorobromoethylene		0.73		
C ₂ HFBr ₄	Tetrabromofluoroethane		0.3-0.8		
C ₂ HF ₂ Br ₃	Tribromodifluoroethane		0.5-1.8		

$C_2HF_3Br_2$	Trifluorodibromoethane		0.4–1.6		
C_2HF_4Br	Tetrafluorobromoethane		0.7–1.2		
$C_2H_2FBr_3$	Tribromofluoroethane		0.1–1.1		
$C_2H_2F_2Br_2$	Difluorodibromoethane		0.2–1.5		
$C_2H_2F_3Br$	Trifluorobromoethane		0.7–1.6		
$C_2H_3FBr_2$	Dibromofluoroethane		0.1–1.7		
$C_2H_3F_2Br$	Difluorobromoethane		0.2–1.1		
C_2H_4FBr	Fluorobromoethane		0.1–0.1		
C_3HFBr_6	Hexabromofluoropropane		0.3–1.5		
$C_3HF_2Br_5$	Pentabromodifluoropropane		0.2–1.9		
$C_3HF_3Br_4$	Tetrabromotrifluoropropane		0.3–1.8		
$C_3HF_4Br_3$	Tetrafluorotribromopropane		0.5–2.2		
$C_3HF_5Br_2$	Pentafluorodibromopropane		0.9–2.0		
C_3HF_6Br	Hexafluorobromopropane		0.7–3.3		
$C_3H_2FBr_5$	Pentabromodifluoropropane		0.1–1.9		
$C_3H_2F_2Br_4$	Tetrabromodifluoropropane		0.2–2.1		
$C_3H_2F_3Br_3$	Trifluorotribromopropane		0.2–5.6		
$C_3H_2F_4Br_2$	Tetrafluorodibromopropane		0.3–7.5		
$C_3H_2F_5Br$	Pentafluorobromopropane		0.9–1.4		
$C_3H_3FBr_4$	Tetrabromofluoropropane		0.1–1.9		

Continued

TABLE 11.2 Class I Ozone-Depleting Substances—Cont'd

Product Name	Chemical Name	Lifetime (years)	ODP	GWP	CAS Number
C ₃ H ₃ F ₂ Br ₃	Tribromodifluoropropane		0.1–3.1		
C ₃ H ₃ F ₃ Br ₂	Trifluorodibromopropane		0.1–2.5		
C ₃ H ₃ F ₄ Br	Tetrafluorobromopropane		0.3–4.4		
C ₃ H ₄ FBr ₃	Tribromofluoropropane		0.0–0.3		
C ₃ H ₄ F ₂ Br ₂	Difluorodibromopropane		0.1–1.0		
C ₃ H ₄ F ₃ Br	Trifluorobromopropane		0.1–0.8		
C ₃ H ₅ FBr ₂	Dibromofluoropropane		0.0–0.4		
C ₃ H ₅ F ₂ Br	Difluorobromopropane		0.1–0.8		
C ₃ H ₆ FBr	Fluorobromopropane		0.0–0.7		

TABLE 11.3 Class II Ozone-Depleting Substances

Product Name	Chemical Name	Lifetime (years)	ODP	GWP	CAS Number
HCFC-21 (CHFCI ₂)	Dichlorofluoromethane	1.7	0.04	143	75-43-4
HCFC-22 (CHF ₂ Cl)	Monochlorodifluoromethane	12	0.05	1780	75-45-6
HCFC-31 (CH ₂ FCI)	Monochlorofluoromethane		0.02		593-70-4
HCFC-121 (C ₂ HFCl ₄)	Tetrachlorofluoroethane		0.01–0.04		354-14-3
HCFC-122 (C ₂ HF ₂ Cl ₃)	Trichlorodifluoroethane		0.02–0.08		354-21-2
HCFC-123 (C ₂ HF ₃ Cl ₂)	Dichlorotrifluoroethane	1.3	0.02	76	306-83-2
HCFC-124 (C ₂ HF ₄ Cl)	Monochlorotetrafluoroethane	5.8	0.02	599	2837-89-0
HCFC-131 (C ₂ H ₂ FCI ₃)	Trichlorofluoroethane		0.007–0.05		359-28-4
HCFC-132b (C ₂ H ₂ F ₂ Cl ₂)	Dichlorodifluoroethane		0.008–0.05		1649-8-7
HCFC-133a (C ₂ H ₂ F ₃ Cl)	Monochlorotrifluoroethane		0.02–0.06		75-88-7
HCFC-141b	Dichlorofluoroethane	9.3	0.12	713	1717-0-6
HCFC-142b (C ₂ H ₃ F ₂ Cl)	Monochlorodifluoroethane	17.9	0.07	2270	75-68-3
HCFC-221 (C ₃ HFCl ₆)	Hexachlorofluoropropane		0.015–0.07		422-26-4
HCFC-222 (C ₃ HF ₂ Cl ₅)	Pentachlorodifluoropropane		0.01–0.09		422-49-1
HCFC-223 (C ₃ HF ₃ Cl ₄)	Tetrachlorotrifluoropropane		0.01–0.08		422-52-6
HCFC-224 (C ₃ HF ₄ Cl ₃)	Trichlorotetrafluoropropane		0.01–0.09		422-54-8
HCFC-225ca (C ₃ HF ₅ Cl ₂)	Dichloropentafluoropropane	1.9	0.02	120	422-56-0

Continued

TABLE 11.3 Class II Ozone-Depleting Substances—Cont'd

Product Name	Chemical Name	Lifetime (years)	ODP	GWP	CAS Number
HCFC-225cb (C ₃ HF ₅ Cl ₂)	Dichloropentafluoropropane	5.8	0.03	586	507-55-1
HCFC-226 (C ₃ HF ₆ Cl)	Monochlorohexafluoropropane		0.02–0.1		431-87-8
HCFC-231 (C ₃ H ₂ FCI ₅)	Pentachlorofluoropropane		0.05–0.09		421-94-3
HCFC-232 (C ₃ H ₂ F ₂ Cl ₄)	Tetrachlorodifluoropropane		0.008–0.1		460-89-9
HCFC-233 (C ₃ H ₂ F ₃ Cl ₃)	Trichlorotrifluoropropane		0.007–0.23		7125-84-0
HCFC-234 (C ₃ H ₂ F ₄ Cl ₂)	Dichlorotetrafluoropropane		0.01–0.28		425-94-5
HCFC-235 (C ₃ H ₂ F ₅ Cl)	Monochloropentafluoropropane		0.03–0.52		460-92-4
HCFC-241 (C ₃ H ₃ FCI ₄)	Tetrachlorofluoropropane		0.004–0.09		666-27-3
HCFC-242 (C ₃ H ₃ F ₂ Cl ₃)	Trichlorodifluoropropane		0.005–0.13		460-63-9
HCFC-243 (C ₃ H ₃ F ₃ Cl ₂)	Dichlorotrifluoropropane		0.007–0.12		460-69-5
HCFC-244 (C ₃ H ₃ F ₄ Cl)	Monochlorotetrafluoropropane		0.009–0.14		134190-50-6
HCFC-251 (C ₃ H ₄ FCI ₃)	Trichlorofluoropropane		0.001–0.01		421-41-0
HCFC-252 (C ₃ H ₄ F ₂ Cl ₂)	Dichlorodifluoropropane		0.005–0.04		819-0-1
HCFC-253 (C ₃ H ₄ F ₃ Cl)	Monochlorotrifluoropropane		0.003–0.03		460-35-5
HCFC-261 (C ₃ H ₅ FCI ₂)	Dichlorofluoropropane		0.002–0.02		420-97-3
HCFC-262 (C ₃ H ₅ F ₂ Cl)	Monochlorodifluoropropane		0.002–0.02		421-2-3
HCFC-271 (C ₃ H ₆ FCI)	Monochlorofluoropropane		+0.001–0.03		430-55-7

TABLE 11.4 Effect of Structure on Properties of Two Different Solvents

Compound	Number of Chlorine Atoms	Number of Fluorine Atoms	Relative Ozone Depletion Potential	Action by U.S. CAA
HCFC 141b	2	1	0.17	Use for cleaning allowed in U.S. through 2003. Banned afterward
CFC-113	3	3	0.8	Production banned in U.S. after 1996
1,1,1 TCA	3	0	0.1	

greater than that of HCFC 141b. So its effect on stratospheric ozone was more significant. *Both emission volume and reactivity were considered in the assignment.*

There is another factor beside chlorine content—survival to reach the stratosphere and survival in the stratosphere. The molecules with the most chlorine are not rated as the worst depleters of ozone. The solvents (chemicals) have to survive in the earth's atmosphere long enough to reach the stratosphere. It is this location where UV radiation is filtered from sunlight by ozone being repeatedly and reversibly converted to oxygen.

- 1,1,1 TCA has an ODP of only 0.1 and three chlorine atoms.
- Perchloroethylene is not listed as an ODC and has four chlorine atoms.

The atmospheric lifetime of perchloroethylene is measured in months; that of CFC-113 (three chlorine atoms) is measured in years.

11.2.2 Reactions of Ozone Depletion in the Stratosphere

Ozone (O_3) is made and destroyed in the stratosphere but generally does not migrate there. It performs a valuable function for the planet by intercepting ultraviolet radiation (UV, wavelength less than about 380 nanometers [380 nm or 3800 Å]) from the sun, and keeps it from impacting us on the earth's surface. Without this blockage, which is a chemical reaction, ultraviolet radiation can cause serious sunburn, skin cancer, and eye damage.

The ozone depletion process begins when CFCs or other ozone-depleting substances are released, usually near ground level. Winds inefficiently mix the troposphere (the closest vertical zone between the earth's surface and about

an altitude of 10 km [6.2 miles]) and evenly distribute the gases. CFCs are extremely stable. They do not dissolve in rain.

After a *period of several years*, CFC molecules reach the stratosphere (the vertical zone between the troposphere and an altitude of about 40 km [25 miles]). Loss of ozone by reaction with CFCs has produced what is popularly (or un-popularly) called the “ozone hole” (see [Figure 11.1](#)).

The following is a simplified description of very complex chemical reactions. The chemical reactions for production of ozone involve two steps.¹²

In the first step, an oxygen molecule absorbs a photon of light (conventionally abbreviated as the Greek character ν with a wavelength shorter than 200 nm). The UV energy splits the oxygen molecule into two oxygen atoms.



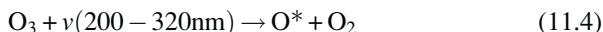
The * symbol on the oxygen atom means that this specie is not a complete molecule and is able to react with other species. The produced oxygen atom (O^*) is very reactive.

In the second step of ozone formation, two oxygen molecules and two oxygen atoms react to form two ozone molecules.

The combined process for ozone formation with UV as an active initiator is



UV light of another wavelength is also involved in the destruction of ozone. UV light whose wavelength is between 280 and 320 nm splits an ozone molecule into an oxygen molecule and an oxygen atom.



The reactive oxygen atom and ozone molecule combine to form two oxygen molecules.



The combined process for ozone destruction, with UV as an active initiator, is the inverse of Eq. (11.3). Ozone is converted to oxygen.



Up to 98% of the sun’s high-energy ultraviolet light is consumed in these chemical reactions involving destruction and formation of atmospheric ozone. The global exchange between ozone and oxygen is on the order of 300 million tons per day.¹³

Because ozone filters out harmful UV radiation, less ozone means higher UV levels at the earth’s surface. Ozone present in the stratosphere protects life on earth by filtering out harmful ultraviolet rays from the sun. The more ozone depletion, the larger is the increase in incoming UV.

UV light would not be blocked from striking the earth if the ozone were somehow eliminated in another chemical process. The process by which ozone is eliminated involves chlorine or bromine atoms from chemicals which include cleaning solvents. Fluorine atoms are too slow to react and do not participate in the ozone depleting reactions. Iodine atoms react too quickly and are consumed at altitudes lower than the stratosphere.

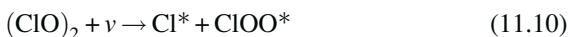
In the stratosphere, CFCs come into contact with short wavelength ultraviolet radiation which is able to split off chlorine atoms from the CFC molecules. The key reaction involves liberation of chlorine or bromine atoms from ODC molecules.¹⁴ The equation is written for CFC-113:



The free chlorine atom (Cl^*) exists for a very short time. If there were no ODCs, the destruction of ozone by the following reactions (11.8–11.11) would not occur. The substance ClO^* also exists for a very short time and is called a free radical.



If this chemical reaction were not followed by three others, the situation would not be so tragic. But three additional reactions regenerate the free chlorine atom:



The result of the last three reactions is that:

- Ozone is converted to oxygen.
- UV radiation is consumed.
- The chlorine radicals are regenerated so they can destroy more ozone. In this process, they are considered to be a catalyst because they are not consumed in the reaction scheme.

The above reactions are not the complete set of all chemical reactions ongoing in the stratosphere. Actual processes are still being researched, and are more complicated than those shown in Eqs. (11.1)–(11.11). The effect of the chlorine radicals can be mitigated via their conversion to hydrochloric acid. Furthermore, the CCl_2F_3 species will also react with chlorine radicals and regenerate the CF-113 .

Because of the slow rate of air mixing between the lower and upper atmosphere, it is theorized that stratospheric CFCs will stay at a significant level well into the next century. CFCs have a lifetime in the atmosphere of about 20–100 years, and consequently one free chlorine atom from a CFC molecule can destroy ozone molecules for a long time.

The Montreal Protocol has had greater impact on the cleaning industry than on the earth's ozone layer, which it was intended to protect. Current progress of ozone concentration change can be tracked at the U.S. Department of Commerce's Climate Monitoring and Diagnostics Laboratory at their website.¹⁶

Many countries in the world have moved to reduce the use of CFCs. The impact of these chemical equations describing the destruction of stratospheric ozone, and the political action they produced, is that chlorofluorocarbon cleaning solvents are no longer produced. Those doing solvent cleaning work are and have been denied their use and must find and have found an alternative.

Regeneration of chlorine radicals challenges users from a philosophical viewpoint. Many feel that their individual choices about chemicals have negligible impact on the global environment. This is because the emissions from a single site are small, relative to the total emissions from this planet. But chlorine atoms in solvents emitted by each user are retained in the stratosphere. They generally are not consumed when they destroy ozone. Choices made by individual users do have consequences after their adoption.^{12,16}

11.2.3 VOC Solvents

Nearly all vaporized chemicals react with UV in sunlight to form ozone in the earth's lower atmosphere (troposphere). They are called Volatile Organic Compounds (VOCs). Examples of VOC cleaning solvents include methyl ethyl ketone (MEK), isopropyl alcohol (IPA), toluene, xylene, aliphatic hydrocarbons, and D-limonene. The byproducts of that reaction form visual smog in the troposphere. This is not ozone depletion, which occurs in the upper atmosphere (stratosphere).

The photochemical reactivity of a few solvents can be low enough that these compounds are defined by the U.S. EPA as not being VOCs. Examples of exempt solvents in the U.S. include *t*-butyl acetate (T-BAC), acetone, methyl acetate (MeOAc), and parachlorobenzotrifluoride (PCFTB).

11.2.3.1 Definition of a VOC

In the U.S., it is the EPA who defines which solvents (chemicals) are and which are not VOCs. This definition is continuously evolving.¹⁷ The definition of VOCs per 40 CFR 51.100(s) is as follows.¹⁸

(s) "Volatile organic compounds (VOC)" means any compound of carbon, excluding carbon monoxide, carbon dioxide, carbonic acid, metallic carbides or carbonates, and ammonium carbonate, which participates in atmospheric photochemical reactions.

1. This includes any such organic compound other than the following, which have been determined to have negligible photochemical reactivity.
2. For purposes of determining compliance with emissions limits, VOC will be measured by the test methods in the approved State implementation

plan (SIP) or 40 CFR Part 60, Appendix A, as applicable. Where such a method also measures compounds with negligible photochemical reactivity, these negligibly-reactive compounds may be excluded as VOC if the amount of such compounds is accurately quantified, and such exclusion is approved by the enforcement authority.

3. As a precondition to excluding these compounds as VOC or at any time thereafter, the enforcement authority may require an owner or operator to provide monitoring or testing methods and results demonstrating, to the satisfaction of the enforcement authority, the amount of negligibly reactive compounds in the source's emissions.
4. For purposes of Federal enforcement for a specific source, the EPA shall use the test methods specified in the applicable EPA-approved SIP, in a permit issued pursuant to a program approved or promulgated under Title V of the Act, or under 40 CFR Part 51, Subpart I or Appendix S, or under 40 CFR Parts 52 or 60. The EPA shall not be bound by any State determination as to appropriate methods for testing or monitoring negligibly reactive compounds if such determination is not reflected in any of the above provisions.

As of this writing, the definition of negligible photochemical reactivity is equivalence with that of ethane. A lower reactivity would be similarly classified as VOC exempt. A greater reactivity would mean no change from VOC status.

There are many uses for solvents in addition to cleaning. VOC exemption of solvents used in these other applications can be quite significant. For example, baking of bread causes emission of copious quantities of ethanol.

Note that evaporation rate is not a defining issue about VOCs in the U.S. The characterization "volatile" is misapplied. Users and suppliers often protest that their chosen solvent should be exempt from VOC status because its rate of evaporation is low. The U.S. EPA assumes that no matter the evaporation rate, all the solvent will eventually reach the troposphere. Hence reactivity in the troposphere with sunlight is the defining issue. Note that this method of regulation of solvents is nearly the reverse of the regulation by the U.S. EPA of solvent emissions from coatings. In the latter case, evaporation rate is a differentiating factor.

VOCs, in the U.S., should be called "PAPRs" (Participates in Atmospheric Photochemical Reactions). That these compounds are volatile and are called organic has no direct relation to the reason for their regulation. The characterization VOC is not relevant, in the U.S. only.

It is just the opposite in Europe. The U.S. definition of VOC includes solvents that are not classified in Europe. The U.S. requirements are more stringent than those of Europe. Ironically, the consequences of the less stringent European VOC definitions make it more difficult to practice solvent cleaning in Europe. In Europe, a VOC is defined by its tendency to evaporate. This type of definition is aimed at reducing emissions from paints, coatings, furniture,

architectural materials, etc. But those doing solvent cleaning are impacted as well.

The most usual definition in Europe is the one stated in the Solvent Emissions Directive for the European Community.^{19,20} The Directive defines a VOC as:

... any organic compound having at 293.15 K (20°C or 68 °F) a vapor pressure of 0.01 kPa (0.075 mmHg) or more, or having a corresponding volatility under the particular conditions of use ...

To make vapor pressure data more useful, use Table 11.5 to convert between units. A number of European countries, however, have developed their own definition in specific contexts.²¹ All are based on vapor pressure or boiling point, not reactivity.

- Germany defines in its 1995 Solvent Ordinance,²² in effect since 25 August 2001, that a VOC has a maximum boiling point of 200 °C (392 °F). Only high boiling (>200 °C) solvents are VOC exempt.
- Switzerland’s VOC Ordinance²³ defines a VOC as an organic compound with a maximum boiling point of 240 °C (464 °F). Only high boiling (>240 °C) solvents are VOC exempt.

The VOC definition in all European countries includes highly volatile solvents which are not considered VOCs in the U.S. (such as above, ethane, *t*-butyl acetate, acetone, methyl acetate, and parachlorobenzotrifluoride). Therefore, the local legislation relative to a specific sector should always be consulted to ensure that the appropriate definition is being used in the context of interest.

These VOC definitions are absolutely crucial for those who wish to clean with solvents and do not wish to emit regulated compounds. In Europe, such operation is limited to solvents which have very low vapor pressure (high boiling point). Consequently, these cleaning solvents:

- Must be heated to a high temperature when boiled in a vapor degreaser. High temperature may cause damage to parts.
- Will impose additional safety hazards because metal surfaces are significantly hotter (100+ °F vs. 300+ °F).

TABLE 11.5 Conversion of Units for Vapor Pressure			
Pa	mbar	mmHg	atm
1	0.01	7.500617×10^{-3}	9.869233×10^{-6}
100	1	0.7500617	9.869233×10^{-4}
133.3224	1.333224	1	1.315789×10^{-3}
101,325	1013.25	760	1

- Will impose thermal stability problems to the solvent because of the increased temperature.
- Will be difficult to remove from parts because of a low evaporation rate.

Differences among nations about definitions of VOC or other environmental parameters are not about to disappear. This is most evident in the different environmental and product safety requirements that exist (or do not exist) in countries and regions throughout the world.

These requirements often stem from fundamentally different perceptions of acceptable risk to human health and the environment. And those differences in turn come from the more basic social, cultural, economic, and even religious attributes of individual countries.

Such differences stubbornly resist easy reconciliation toward global uniformity. Also there are legal/regulatory precedents. The stability of precedent is one of the most valuable aspects of any legal/regulatory system. It assures that existing rules will not change readily so the public can confidently draft and sign contracts, agree to specifications, etc., on the basis of clear principles. Equally important, it allows for reasonable predictions of changes. Certainly, change to a global system of environmental regulation is not likely to occur soon.

Another approach to VOC regulation (exemption) has been taken by the International Standards Organization (ISO) in their ISO 16000-6.²⁴ Here, vapor concentrations are measured via sampling in a work area or a test chamber. The sample is collected on a sorbent and analyzed with a gas chromatograph. This approach mainly applies to emissions not from surface cleaning machines but from products. The regulating agency²⁵ then specifies which solvents are of concern.

11.2.3.2 *Definition of VOC Exempt*

The list of chemicals currently exempt in the U.S. is shown in [Table 11.6](#). In addition to those chemicals already exempted, there are additional compounds for which their manufacturer, distributor, or trade association has filed for exempt status in the U.S. EPA. These are listed in [Table 11.7](#) in order of earliest application date (Received Petitions Requesting VOC Exempt Status and for which EPA has Published no Final Action as of October 22, 2002). As this chapter is written, the U.S. EPA has proposed to grant VOC exemption to the four chemicals in [Table 11.7](#) whose names are printed in boldface.²⁶

In a sense the U.S. EPA's policy of VOC exemption has not succeeded. The ideal policy would:

- Classify solvents as VOCs by their potential for smog formation. The current policy does not do this. The classification scheme is by negligible reactivity and all else.

TABLE 11.6 Compounds Exempt from VOC Status in the U.S.	
Chemical Name	Industry Name
Methane	
Ethane	
Methylene chloride	Dichloromethane
1,1,1-Trichloroethane	Methyl chloroform, TCA
1,1,2-Trichloro-1,2,2-trifluoroethane	CFC-113
Trichlorofluoromethane	CFC-11
Dichlorodifluoromethane	CFC-12
Chlorodifluoromethane	HCFC-22
Trifluoromethane	HFC-23
1,2-Dichloro 1,1,2,2-tetrafluoroethane	CFC-114
Chloropentafluoroethane	CFC-115
1,1,1-Trifluoro 2,2-dichloroethane	HCFC-123
1,1,1,2-Tetrafluoroethane	HFC-134a
1,1-Dichloro 1-fluoroethane	HCFC-141b
1-Chloro 1,1-difluoroethane	HCFC-142b
2-Chloro-1,1,1,2-tetrafluoroethane	HCFC-124
Pentafluoroethane	HFC-125
1,1,2,2-Tetrafluoroethane	HFC-134
1,1,1-Trifluoroethane	HFC-143a
1,1-Difluoroethane	HFC-152a
Parachlorobenzotrifluoride	PCBTf
Cyclic, branched, or linear completely methylated siloxanes	OS-10, OS-20, OS-30
Acetone	
Tetrachloroethylene	Perchloroethylene
3,3-Dichloro-1,1,1,2,2-pentafluoropropane	HCFC-225 ca
1,3-Dichloro-1,1,2,2,3-pentafluoropropane	HCFC-225 cb
1,1,1,2,3,4,4,5,5,5-decafluoropentane	
Difluoromethane	HFC-32

Continued

TABLE 11.6 Compounds Exempt from VOC Status in the U.S.—Cont'd

Chemical Name	Industry Name
Ethyl fluoride	HFC-161
1,1,1,3,3,3-Hexafluoropropane	HFC-236 fa
1,1,2,2,3-Pentafluoropropane	HFC-245 ca
1,1,2,3,3-Pentafluoropropane	HFC-245 ea
1,1,1,2,3-Pentafluoropropane	HFC-245 eb
1,1,1,3,3-Pentafluoropropane	HFC-245 fa
1,1,1,2,3,3-Hexafluoropropane	HFC-236 ea
1,1,1,3,3-Pentafluorobutane	HFC-365 mfc
Chlorofluoromethane	HCFC-31
1-Chloro-1-fluoroethane	HCFC-151 a
1,2-Dichloro-1,1,2-trifluoroethane	HCFC-123 a
1,1,1,2,2,3,3,4,4-Nonafluoro-4-methoxy-butane	C ₄ F ₉ OCH ₃
2-Difluoromethoxymethyl-1,1,1,2,3,3,3-heptafluoropropane	(CF ₃) ₂ CFCF ₂ OCH ₃
1-Ethoxy-1,1,2,2,3,3,4,4,4-nonafluorobutane	C ₄ F ₉ OC ₂ H ₅
2-Ethoxydifluoromethyl-1,1,1,2,3,3,3-heptafluoropropane	(CF ₃) ₂ CFCF ₂ OC ₂ H ₅
Methyl acetate	Polymerization
Perfluorocarbon compounds which fall into these classes: (I) Cyclic, Branched, or Linear, Completely Fluorinated Alkanes. (II) Cyclic, Branched, or Linear, Completely Fluorinated Ethers with no Unsaturations	

- Encourage substitution of lower reactivity smog-formers for higher reactivity smog-formers. The current policy provides little incentive for solvent substitution. This is because it is a binary policy—a chemical is either a VOC or not a VOC. There is no recognition of higher reactivity or lower reactivity. Today, there is considerable scientific discussion about a universal reactivity characteristic. Thus, solvent substitution (except for that mandated by the Montreal Protocol) has not occurred in the cleaning industry to a significant degree.

In addition, use of the per-unit-weight basis is inconsistent with the selection of ethane as the reactivity benchmark. It creates a bias that causes reactive, high molecular weight organics to be classified as negligibly reactive.²⁷ Yet, the EPA has had inconsistent views about the reactivity benchmark—acetone was excepted on a per-mole basis.

TABLE 11.7 Chemicals for Which VOC Exempt Status Has Been Applied

Chemical Name	Petitioner
1-Bromopropane (also known as <i>n</i> -propyl bromide)	Enviro Tech International, Alameda, CA (submitted May 10, 1996). Petition also submitted by Albemarle Corp., Baton Rouge, LA (submitted November 18, 1997)
Methyl bromide	Chemical Manufacturers Association, Washington, DC (submitted July 19, 1996)
<i>n</i> -Alkanes (C12–C18)	The Aluminum Association, Washington, DC (submitted November 27, 1996)
Technical white oils	The Printing Industries of America and Pennzoil Products Company (submitted December 20, 1996)
<i>t</i> -Butyl Acetate	ARCO Chemical Company [now Lyondell] (submitted January 17, 1997) [Proposed September 30, 1999, see 64 FR 52731]
Benzotrifluoride	Occidental Chemical Company, Niagara Falls, NY (submitted March 11, 1997)
Carbonyl sulfide (COS)	E.I. du Pont de Nemours and Company (submitted August, 11, 1997). Petition also submitted by Texas Mid-Continent Oil & Gas Association (submitted December 5, 1997)
Trans-1,2-dichloroethylene	3 M Corporation, St. Paul, MN (submitted October 8, 1997)
Dimethyl succinate	Dibasic Esters Group, affiliated with the Synthetic Organic Chemical Manufacturers Association, Inc. (submitted October 14, 1997)
Dimethyl glutarate	Dibasic Esters Group, affiliated with the Synthetic Organic Chemical Manufacturers Association, Inc. (submitted October 14, 1997). Submitted in the same petition as for dimethyl succinate
Carbon disulfide	Texas Mid-Continent Oil & Gas Association (submitted December 5, 1997)
Acetonitrile	BP Chemicals and GNI Chemicals Corporation (submitted January 21, 1998)
Toluene diisocyanate (TDI)	Chemical Manufacturers Association (The Diisocyanate Panel of CMA reported the following members: ARCO Chemical Company, BASF Corporation, Bayer Corporation, The Dow Chemical Company, and ICI Americas, Inc.), submitted January 22, 1998

Continued

TABLE 11.7 Chemicals for Which VOC Exempt Status Has Been Applied—Cont'd

Chemical Name	Petitioner
HFC-227ea (1,1,1,2,3,3,3-heptafluoropropane)	Great Lakes Chemical Corporation (submitted February 18, 1998)
Methylene diphenyl diisocyanate (MDI)	Chemical Manufacturers Association (The Diisocyanate Panel of CMA reported the following members: BASF Corporation, Bayer Corporation, The Dow Chemical Company, ICI Americas, Inc., and Lyondell Chemical Company), submitted August 19, 1998
1,1,1,2,2,3,3-Heptafluoro-3-methoxy-propane ($n\text{-C}_3\text{F}_7\text{OCH}_3$)	3 M Performance Chemicals and Fluids Division (submitted February 5, 1999)
Propylene carbonate	Huntsman Corporation, Austin, Texas (July 27, 1999)
Methyl pivalate	Exxon Chemical Company, Houston, TX (November 22, 1999)
3-Ethoxy-1,1,1,2,3,4,4,5,5,6,6-dodecafluoro-2-(trifluoromethyl) hexane (also known as HFE-7500 or L-15381)	3 M Corporation, St. Paul, MN (August 21, 2000)
Methyl formate	Foam Supplies, Inc., Earth City, Missouri (February 12, 2002)

As described above, legislation in the U.S. and Europe is different and not connected. A product classified as a non-VOC in the U.S. does not automatically receive the same classification in Europe. There are no specific exemptions in Europe from VOC status.

11.2.3.3 Reactions Leading to Smog Formation

VOCs can react with emissions from cars and diesel engines to cause air pollution problems in some areas.²⁸ VOCs are regulated because they react with sunlight and other chemicals in the atmosphere to produce what we know as photochemical smog.

Note that VOCs as emitted chemicals do not produce smog. They may add an odor, a texture, or a color to air. But they do not form smog without the presence of other pollutants.

Smog is produced by a complex photochemical reaction between hydrocarbons and nitrogen oxides in the presence of sunlight.

- Smog (a.k.a. “photochemical smog”) can be formed just from nitrogen oxides and sunlight—without presence of VOC. This smog is chiefly nitrogen oxides (NO_x) and ozone (O_3).
- Smog cannot be formed from just VOCs and sunlight; oxides of nitrogen and an oxidizer (ozone) are required in the chemical reactions. Essentially, the contribution of solvent emissions (VOC) is to make higher existing levels of photochemical smog worse.

The oxides of nitrogen generally come from combustion processes, chiefly gasoline-powered automobiles, forest fires, and fuel-burning power plants. That is why cars have catalytic converters and power plant stacks have scrubbers. The catalytic converter in automobile exhaust systems reduces air pollution by oxidizing hydrocarbons to CO_2 and H_2O and, to a lesser extent, converting nitrogen oxides to N_2 and O_2 . Oxides of nitrogen are often called NO_x , meaning that NO and NO_2 are included.

11.2.3.4 Smog Formed from VOCs

In bright sunlight, nitrogen oxides, hydrocarbons (VOC, also called HCs), and oxygen interact chemically to produce powerful oxidants like ozone (O_3), and peroxyacetyl nitrates (PANs). Ozone and PANs are temporarily created, but are destroyed in subsequent reactions. PANs are primarily responsible for the eye irritation so characteristic of this type of smog. Both pollutants have significant effect on human health.

The path to smog formed from VOCs starts with the reactive oxidizers (O_3 and O^*) available from the chemistry associated with pollution byproducts from combustion.

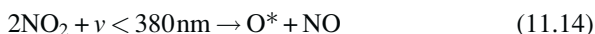
The first reaction is simple: combustion (oxidation) which occurs on the earth’s surface.



The second reaction involves further oxidation of a combustion product (nitric oxide) to nitrogen dioxide. This oxidation occurs in the atmosphere.



The third reaction occurs in the troposphere. Nitrogen dioxide degrades under the influence of UV light. The reactive O^* radical is produced.



O^* (a reactive radical) reacts with water to produce HO^* —another radical. The HO^* specie (called a hydroxy radical) is unstable and then quickly reacts with hydrocarbons.

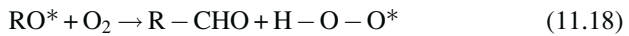


The specie R^* is called a hydrocarbon radical. Like other radicals, it is very reactive. (“R” is the HC on to which the aldehyde is attached.)



The specie ROO^* is called a peroxide radical. A peroxide is a pair of linked oxygen atoms attached to a HC. This specie exists because there are free oxygen atoms (O^*) which produce its precursors R^* and HO^* . And it does not exist for very long.

The peroxide radical reacts next, in two steps, with pollution from combustion (NO) and oxygen to make a stable compound—an aldehyde (CHO).

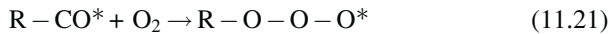
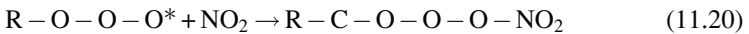
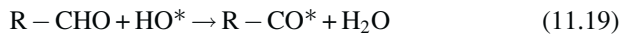


Note that Eq. (11.17) is the first in which pollution byproducts of combustion and VOCs interact.

An aldehyde is a compound which contains the group CHO. The group HOO is called a hydro peroxide (see Figures 11.2 and 11.3).

Aldehydes and peroxyacetyl nitrate (PAN) are a common component in analyses of air above Mexico City, Santiago de Chile, Hong Kong, and other cities affected with pollution from combustion and hydrocarbons.²⁹

The formation of PAN (see Figure 11.4) proceeds from aldehydes produced as above. Three rapid steps are involved:



For the case where R is C_4 (methane), the product in Eq. (11.18) is PAN. PAN causes eye irritation, chest constriction, irritation of mucous membranes, and is also known to damage plants.

The hydro peroxide formed by reaction (11.18) is also a reactant which can produce byproducts hazardous to humans and the earth.

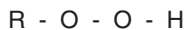
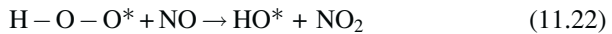


FIGURE 11.2 Structure of peroxide.

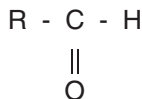


FIGURE 11.3 Structure of aldehyde.

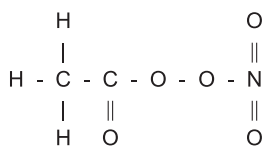


FIGURE 11.4 Structure of PAN (peroxyacetyl nitrate).

In summary, the sequence of reactions associated with smog formation is as follows:

- Nitrogen oxides result from combustion processes and generate oxygen atoms and ozone.
- Oxygen atoms form hydroxy radicals which generate hydrocarbon radicals.
- Hydrocarbon radicals form hydrocarbon peroxides which generate aldehydes.
- Aldehydes form aldehyde peroxides and hydro peroxides which generate peroxyacetyl nitrates and nitric acid.

The concern about these reactants, intermediates, and products is deadly serious (see [Table 11.8](#)).

Hydrocarbon solvents are not the prime cause of contamination of the earth’s air. There would be no formation of smog unless the influence of hydrocarbon solvent emissions was combined with the emissions from combustion processes.

TABLE 11.8 Major Pollutants and Reaction Products			
Pollutant	Description	Effects	Levels of Concern
Nitric oxide (NO)	Nontoxic, colorless, odorless gas	Easy to oxidize to NO ₂	
Nitrogen dioxide (NO ₂)	Toxic gas	Gives polluted air a yellow to reddish brown color and pungent odor	Hundreds of ppb
Ozone	Active oxidant	Is most destructive to plants	Hundreds of ppb or less
PAN		Inflammation of air passages, reducing lung capacity. Long-term damage to lungs and ability to resist other lung and throat problems. Cause a plant disorder known as “silver leaf”	10–20 ppb

The equations and structures in this section may appear complicated to a user trying to remove oil and metal particles from machined injection valves. Yet these equations and structures are important. They are the scientific basis for regulations defining which solvents can be used to remove that soil on parts, how the solvent cleaning equipment must be designed, and how employees using that equipment must be protected.

11.2.4 Global Warming

The term “global warming” is a synonym for massive climate change on our planet. The concern is that activities of man have affected the earth’s atmospheric environment to the extent that the earth’s climate will be affected.

The pace of global warming is glacial. The nature of global warming is that the incremental changes of which it is composed are minuscule. That is why it is so difficult to recognize what it is, and what its causes are.

The Earth naturally absorbs incoming solar radiation. It also reflects and emits longer wavelength terrestrial (thermal) radiation back into space. On average, the absorbed solar radiation is balanced by the outgoing terrestrial radiation emitted to space—atmospheric temperatures are stable.

A portion of this terrestrial radiation is absorbed by gases in the atmosphere. The energy from this absorbed terrestrial radiation warms the earth’s surface and atmosphere. This creates what is known as a “natural greenhouse effect.”

The essence of global warming is that if gases in the upper atmosphere have a higher capacity to absorb energy, they will do so and the earth’s surface will become warmer.

11.2.4.1 Regulation of Solvent Cleaning Because of Global Warming

Global Warming Potential (GWP) has been developed as a metric to compare relative to another gas the ability of each greenhouse gas to trap heat in the atmosphere. Carbon dioxide (CO₂) was chosen as the reference gas to be consistent with the guidelines of the Intergovernmental Panel on Climate Change (IPCC). The IPCC was created jointly by the World Meteorological Organisation and the United Nations Environment Programme in 1988. The IPCC is responsible for compiling and synthesizing the growing body of scientific literature on climate change. The comprehensive assessments of IPCC form the scientific basis for climate change policies.

GWP of common solvents, which are important to persons doing solvent cleaning, are given in Table 11.9 with that of other compounds of interest.³⁰ They are sorted by GWP.

The gases of concern in Table 11.9 are the HFCs and PFCs. They are called high-GWP gases. By application, most of these solvents are refrigerants or blowing agents for foams.

TABLE 11.9 Global Warming Potential (GWP) of Certain Chemicals				
Gas	Formula	Source	Application	100 Year GWP
Carbon dioxide	CO ₂	Natural		1
Methane	CH ₄			21
Nitrous oxide	N ₂ O			310
HFE-7200	C ₄ H ₉ OCH ₃	Manmade	Cleaning agent	320
HCFC-225ca/ HCFC-225cb	CF ₃ CF ₂ CHCl ₂ CClF ₂ CF ₂ CHClF		Cleaning and drying agent	181/620
HFC-365 mfc	CF ₃ CH ₂ CF ₂ CH ₃		Cleaning and drying agent	890
HFC-4310 mee	CF ₃ CHFCHFCF ₂ CF ₃		Cleaning and drying agent	1500
HFC-227 ea	CF ₃ CHFCF ₃		Fire extinguishent Refrigerant	2900
CFC-113	CCl ₂ FCClF ₂		Cleaning agent whose manufacture is banned because of ODP	6000
HFC-236 fa	CF ₃ CF ₂ CF ₃		Fire extinguishent refrigerant	6300
PFC C ₃ F ₈	C ₃ F ₈		Drying agent heat transfer fluid	8600
Sulfur hexafluoride	SF ₆		Etchant in manufacture of semiconductors	22,200

Even on the basis of carbon equivalents (units of millions of metric tons of carbon equivalents [MMTCE], not actual volume emitted), these gases with high-GWP ratings are not a major contributor to the inventory of greenhouse gases.³¹ No single cleaning solvent is a major contributor to the problem of global warming—based on GWP rating and emission volume.

Use of HFCs, PFCs, and SF₆ draws concern from environmental regulatory agencies, out of proportion to their volume (or carbon-equivalent volume) of emission. There are several reasons for a high level of concern by environmental regulatory agencies:

- Use is controllable with moderate regulatory effort. Contrast regulation of CO₂ emissions from human respiration or photosynthesis in plant life; regulation of CH₄ emissions from animals; regulation of N₂O emissions from combustion of forests, with regulation of use of a chemical whose production can be restricted (or banned) by fiat, or whose use can be managed through choices of process equipment. With this perspective, focus on high-GWP gases is a sound regulatory strategy.
- Use of fluorinated gases is growing at an aggressive rate. That rate is expected to accelerate faster than general economic growth.³² The HFC category is expected to grow from less than 10 MMTCE to more than 50 in 20 years. A major reason for growth of use of HFCs and PFCs is that they replace other substances which are CFCs (CFC-113 and HCFC 141b).
- Regulation is consistent with the economic preferences of users. HFCs and PFC are quite expensive (U.S. \$15-US\$20 per pound as of fall 2003). Users have a real incentive not to emit them. Most applications, except for actions such as suppression of fires, are in equipment which contains emissions.

Fluorinated gases have higher GWP values than other substances (see Table 11.8). Their use commands attention.

11.2.4.2 *Specific Regulations Affecting Solvent Cleaning*

HCFCs contain carbon, hydrogen, chlorine, and fluorine. They are regulated by (at least) the Montreal Protocol.

HFCs contain no chlorine. They are not regulated by the Montreal Protocol because they do not react to deplete ozone from the stratosphere. EPA has accepted them as substitutes for CFC. For another point of view about use of HFCs as replacements for ODCs, see Ref. 33. A quotation from that source is "... Clearly, HFCs cannot be the solution to ozone depletion. They merely replace the ozone crisis with yet another environmental disaster...."

Though EPA has accepted HFCs as substitutes for CFC-113, they may be regulated in the future³⁴ because of their use volume and GWP (for example, the GWP for HFC-4310 mee is 1300).

Under the Montreal Protocol, the U.S. and other developed nations are obligated to achieve a certain percentage of progress toward the total phaseout of HCFCs, by certain dates. These nations use a cap as a baseline to measure their progress toward achieving these percentage goals.

Within the U.S., there is a schedule for phasing out its use of HCFCs in accordance with the terms of the Montreal Protocol. The U.S. EPA plans for the U.S. to meet the limits set under the Protocol by accelerating the phase out of HCFC-141b, HCFC-142b, and HCFC-22. These are the most damaging of the HCFCs. Comparison of the Montreal Protocol and U.S. phase out schedules is given in Table 11.10.

The Kyoto Protocol, adopted in the late 1990s about controlling the pace of increase of global warming, is an agreement made under the United Nations Framework Convention on Climate Change (UNFCCC). Under the Kyoto

TABLE 11.10 Bans Associated with HCFC Cleaning Solvents			
Montreal Protocol		United States	
<i>Year to be Implemented</i>	<i>Reduction in Consumption, Using the Cap as a Baseline (%)</i>	<i>Year to be Implemented</i>	<i>Implementation of HCFC Phase-out through Clean Air Act Regulations</i>
2004	35.0	2003	No production and no import of HCFC-141b (already implemented)
2010	65.0	2010	No production and no import of HCFC-142b and HCFC-22, except for use in equipment manufactured before January 1, 2010 (so no production or import for NEW equipment that uses these refrigerants)
2015	90.0	2015	No production and no import of any HCFCs, except for use as refrigerants in equipment manufactured before January 1, 2020. This specifically impacts those doing solvent cleaning work with HCFC 225
2020	99.5	2020	No production and no import of HCFC-142b and HCFC-22
2030	100	2030	No production and no import of any HCFCs

Protocol, there are six gases that count toward an emissions reduction target. These gases are CO₂, CH₄, N₂O, HFC, PFC, and SF₆. These last three are man-made gases, sometimes called “potent industrial greenhouse gases,” “greenhouse gases,” or “fluorinated greenhouse gases (f-gases).”

Some countries have taken an aggressive position about use of HFCs for all but a few limited use applications for which an exemption has been made. Solvent cleaning is not exempt from these bans. Countries which have or are debating prohibition of HFC use include Iceland, Denmark (planned for 2006), Austria (proposed for 2008), Germany, Switzerland, Sweden, and Belgium.

TABLE 11.11 Atmospheric Concerns About Common Cleaning Solvents

Solvent	ODP	GWP	VOC	Regulatory Status
Acetone	NA	NA	No	OK
HFE-7100	0	320	No	OK
HFC-365 mfc	0	890	No	OK
HFC 43-10 mee	0	1300	No	OK
TCE	0	0	Yes	OK
<i>n</i> -Propyl Bromide	0.026	0	Yes	Recently approved by EPA's snap ³⁷
IPA	NA	NA	Yes	OK
HCFC 225 ca/cb	0.03	180/620	No	Phase-out in 1015
HCFC 141 b	0.12	713	No	Already phased out
1,1,1 TCA	0.1	140	Yes	
CFC-113	1.0	6000	No	

Other countries (France, United Kingdom, The Netherlands, and Ireland) seek to limit use and emissions of HFCs via taxes on purchase. The Environmental Ministry of Japan has announced their intent to encourage Japanese business and industry to seek alternatives to ozone-depleting and global warming refrigerants.

The characterization of atmospheric chemistry for various common cleaning solvents is summarized in [Table 11.11](#).^{35,36}

The limitation within [Table 11.11](#) is that it addresses only one issue in the minds of prospective users of cleaning solvents—regulations associated with chemical reactions in the earth's atmosphere. Other issues (performance, cost, safety, odor, etc.), covered elsewhere in this chapter, are not included.

From the information in [Table 11.11](#), HFE-7100 might be the singular choice by all users (and indeed it might well be for some applications). However, for reasons discussed in other sections of this chapter, some users do make other choices. One of those reasons is that it is nearly useless for soils containing significant polarity or hydrogen bonding.

11.2.5 Relationship of Solvent Characteristics ODP, VOC, and GWP

Three atmospheric chemical reactions of solvents are important to users and draw concern from regulators. Together they add meat to the “alphabet soup” of jargon which complicates our language: ODP, VOC, and GWP.

- The “Ozone Hole” over Antarctica is caused by ozone-depleting compounds (ODCs) reacting with sunlight in the stratosphere to consume ozone. These compounds are rated by their potential to so react by their Ozone Depletion Potential (ODP).
- Photochemical smog is generally caused by Volatile Organic Compounds reacting with sunlight, and combustion products, in the troposphere.
- Global Warming Potential is an attempt to predict the effect a stable, long-lived molecule might have in warming the atmosphere of the planet.

Can one solvent have all these three characteristics? It is not likely. Two of these three characteristics imply contradictory behavior. But two of these three do have similar behavior.

Many common chemicals have GWP ratings. In general, chemicals that have high GWP ratings have low VOC ratings. High VOC ratings generally mean a chemical has a low GWP. It is almost impossible for a chemical to be both a low VOC and a low GWP. The reason is that:

- A chemical which has a high GWP rating must survive passage, without decomposition, through the troposphere and penetrate the stratosphere. Said another way, a high GWP rated gas is one which failed to react with UV in the troposphere. That is what a VOC does.
- A chemical which has a high VOC rating, or at least is not exempt by the U.S. definition, will react with UV in the troposphere and not survive to penetrate the stratosphere. Stated another way, VOCs are not stable long-lived molecules as are chemicals with a high GWP rating.

Chemicals with high GWP ratings absorb thermal energy (heat). Chemicals with high ODP react with sunlight (radiation). While not the same, both characteristics are similar.

To summarize:

- ✓ ODP and GWP are somewhat related.
- × VOC and ODP do not correlate.
- × VOC and GWP do not correlate.

11.3 POTENTIAL HEALTH CONSEQUENCES OF SOLVENT USE IN CLEANING

It would be wrong to say:

- all solvents are hazardous;
- some solvents are hazardous, and some are not; or
- all solvents are not hazardous.

However, it would be correct to say that all chemicals present some hazards. Learning to manage them is a more fruitful approach than avoiding their use entirely or embracing them for every application.

The title of this section starts with the word *potential*. The key to completing cleaning work in a safe and effective manner is to keep that potential from becoming a reality.

This section will include descriptions of the following general hazards:

- Fire, as indicated by flash point testing, flammability testing, and evaluation of electrostatic discharge.
- Body contact, achieved through inhalation, ingestion, and/or skin contact.
- Carcinogenicity.

This section will also include descriptions of the following methods for avoiding or coping with these hazards:

- Obtaining the proper information about the specific hazards associated with individual solvents.
- Taking precautions to avoid the impact of hazards by using protection equipment and implementing procedures.

11.3.1 Flammability Issues

Many solvent suppliers claim that their solvent cannot be ignited or catch fire. What they really mean is that their solvent product has no *flash point*. There is a discontinuity in communication when the technical definition of flammability is used because users' definitions of flammability may differ from the technical definition.

The technical definition is the basis for many regulations and industrial practices pertaining to cleaning solvents. Solvents are classified as flammable (value < 200 °F) based solely on flash point data (classification published in Occupational Safety and Health Administration (OSHA) Standard 1910.106³⁸). The definition commonly used and understood for flammability is that the described product will not burn. Definitions for “flammability” and “flammable” in various non-technical dictionaries include: “... measure of the extent to which a material will support combustion ...,” “... any substance that is easily ignited, burns intensely, or has a rapid rate of flame spread ...,” “... capable of being easily ignited and of burning quickly ...,” etc.

Some suppliers claim their products that do not have measured flash points are not flammable. Users believe this statement means that these products cannot catch fire. Very often, the statement is not true. This situation arises because flash point data for halogenated solvents can be misleading. Confusion exists because halogenated solvents contain both fuel (hydrogen and carbon atoms) and a fire suppressant (the halogen atom). Which one dominates depends on various factors.

11.3.1.1 Flash Point

The flash point of a solvent has no direct effect on its capability as a cleaning solvent. It does, however, have a dominant indirect effect on selection

of the cleaning process, cleaning equipment, and cleaning procedures, as well as packaging, transportation, selection, and disposal. In other words, specifications for the above issues all incorporate flash point. *Other than solvency, there is no more important parameter than flash point in solvent cleaning.*

“Flash point” is the minimum temperature at which a liquid gives off vapor within a test vessel in sufficient concentration to form an ignitable mixture with the air near the surface of the liquid. The lower the flash point, the easier it is to ignite a liquid solvent. Specific details about how to measure flash point are found in OSHA Standard 1910.106.³⁸

Measurement of flash point does not require production of a stable flame, just ignition of vapor and fuel. The actual measurement which defines ignition is usually a temperature rise produced by the combustion.

11.3.1.1.1 Flash Point Test Equipment

Flash point test equipment has been designed to simulate the various ways in which solvents are used. Four types are described in Table 11.12. One type of equipment is shown in Figure 11.5.

The majority of solvent cleaning work is performed in equipment of two types relative to flash point, either an *open tank* or a *closed tank*.

An *open tank* would typically be used for cleaning at less than the boiling point (cold cleaning). The potential for ignition of solvents in cold cleaning, under shipment, or storage conditions is assessed by measuring flash point in an *open cup* tester.

A *closed tank* would include a storage or shipping container, or an enclosed machine. In this sense, a vapor degreaser is a closed tank. The potential for ignition of closed tank solvents under cold cleaning conditions is assessed by measuring flash point in a *closed cup* tester.

Flash points are determined experimentally by heating the liquid in a container (cup) and then introducing a small flame just above the liquid surface. The temperature at which there is a flash/ignition is recorded as the flash point.

Obviously, the same solvent will give different absolute results, and may give different results relative to another solvent, if tested in an open cup or a closed cup tester. The closed-cup method prevents vapors from escaping. The open cup tester, on the other hand, will lose the most volatile components. Thus, open-cup flash points are higher than those for the same solvent measured in the closed cup tester. When several values are available, the lowest temperature is usually taken in order to assure safe operation of the process.

Do not ever accept flash point information if the type of tester is not specified as well. Flash points for cleaning solvents are normally measured by the TAG closed cup test (named for Giuseppe Tagliabue, who developed it). The TAG test is also referred to as “Tag CC,” “TAG CC,” or simply “CC,” for “closed cup.”

TABLE 11.12 Comparison of Flash Point Test Equipment

Name	Test Fluids	Temperature Range (°C)	Cup Open/Closed	Detection Method	ASTM Test	Ignition Source
Tag	Cleaning solvents, etc.	−20 to +80	Open/closed	Temperature rise	D1310 ³⁹ (open), D56-02 ⁴⁰ (closed)	Gas or electric igniter
Setaflash	Diesel, kerosene, lacquers	0–220	Open/closed	Visual	D3278–82 (closed)	
Pensky-Martens (shown in Figure 11.5)	Paints, coatings	60–360	Closed	Temperature rise	D 93	
Cleveland	Oils, bitumens	>360	Closed	Ionization detection	D 92	



FIGURE 11.5 Pensa–Martens closed cup tester with mixer for determining the flash point of viscous paints.

11.3.1.1.2 Regulatory Requirements Related to Flash Point

Three U.S. government agencies and one private agency have requirements for users of cleaning solvents, based on the flash point of the solvent used. Again, these requirements have at least as much effect on the choice of cleaning solvent as does its solvency power.

These requirements are defined in a classification system which is published in OSHA Standard 1910.106.³⁸

- Class IA—Flash Point less than 73 °F (22.7 °C); Boiling Point less than 100 °F (37.8 °C)
- Class IB—Flash Point less than 73 °F; Boiling Point equal to or greater than 100 °F
- Class IC—Flash Point equal to or greater than 73 °F, but less than 100 °F
- Class II—Flash Point equal to or greater than 100 °F, but less than 140 °F (60 °C)
- Class IIIA—Flash Point equal to or greater than 140 °F, but less than 200 °F (93.3 °C)
- Class IIIB—Flash Point equal to or greater than 200 °F

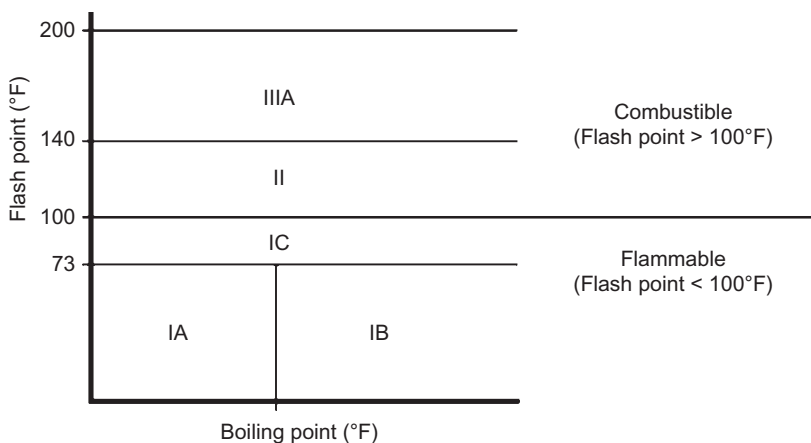


FIGURE 11.6 Classes of flammable and combustible liquids as defined in 29 CFR-1910.106.³⁸

This information is collected in [Figure 11.6](#).

Note that boiling point is used only to distinguish between Class IA and Class IB solvents. Class IA liquids are extremely volatile, but there are few liquids that are so classed.

Theoretically, there is no upper limit to Class IIIB, except that liquids with a closed cup flash point above 200 ° F dry slowly and are poor choices for cleaning solvents used in vapor degreasing operations.

The U.S. Department of Transportation (DOT) has a classification system that is only slightly different. Because they are partners in a worldwide network of regulations about hazardous materials, DOT has changed its definition of “flammable liquid” by raising the upper limit to 141 °F (60.5 °C). However, DOT regulations include a so-called “domestic exemption” that allows a shipper to redesignate as a combustible solvent any solvent whose flash point is in the NFPA Class II range and which does not meet any other hazardous material definition.

Note that this classification applies to all liquids, not just liquids used for cleaning, and that the closed cup method is used for all determinations of flash point.

11.3.1.1.3 Flash Point Data

Data from a variety of non-halogenated (i.e. not containing fluorine, bromine, or chlorine atoms) solvents are shown in [Table 11.13](#).⁴¹ Solvents commonly used for cleaning are printed in boldface type. Data are sorted by flash point value and the National Fire Protection Association (NFPA) classification is differentiated by the shaded background.

TABLE 11.13 Flash Point Data for Solvents

Chemical	Flash Point		Boiling Point		NFPA Class
	° F	°C	°F	°C	
Propane	(157)	(105)	(44)	(42)	IA
Pentane	(57)	(49)	97	36	IA
Ethyl ether	(49)	(45)	95	35	IA
Acetaldehyde	(38)	(39)	69	21	IA
Dimethyl sulfide	(36)	(38)	99	37	IA
Carbon disulfide	(22)	(30)	115	46	IB
Ethylene oxide	(20)	(29)	55	13	IA
<i>n</i> -Hexane	(7)	(22)	156	69	IB
Acetone	(4)	(20)	56	133	IA
Cyclohexane	(4)	(20)	179	81	IB
Tetrahydrofuran	6	(14)	153	67	IB
Benzene	12	(11)	176	80	IB
Triethylamine	20	(7)	193	89	IB
Methyl ethyl ketone (MEK)	25	(4)	176	80	IB
Toluene	40	4	231	111	IB
Methyl alcohol	52	11	149	65	IB
Isopropyl alcohol (IPA)	53	12	180	82	IB
Ethyl alcohol	55	13	173	78	IB
Pyridine	68	20	239–241	116	IB
2-Nitropropane	75	24	248	120	IC
Tert butyl isocyanate	80	27	185–187	85–86	IC
Chlorobenzene	82	28	270	132	IC
Epichlorohydrin	88	31	239–243	115–117	IC
Xylene	81–90	27–32	280–291	138–144	IC
Morpholine	100	38	263	128	II
Acetic acid, glacial	103	39	244	48	II
Bromobenzene	118	48	307–316	153–158	II
Formic acid	122	50	213	101	II

Continued

TABLE 11.13 Flash Point Data for Solvents—Cont'd

Chemical	Flash Point		Boiling Point		NFPA Class
	° F	°C	°F	°C	
Methyl lactate	135	57	291	144	II
Stoddard solvent	100–140	38–60	300–400	150–200	II
<i>Iso</i> -propyl lactate	140	60	315	157	II
Ethyl lactate	142	61	307	153	IIIA
Benzaldehyde	145	63	352	178	IIIA
Cyclohexanol	154	68	322	161	IIIA
Tetrahydronaphthalene	160	71	406	208	IIIA
<i>Iso</i> -butyl lactate	169	76	360	182	IIIB
Methacrylic acid	170	77	316	158	IIIA
Butyl lactate	174	79	369	187	IIIB
Nitrobenzene	190	88	412	211	IIIA
<i>n</i> -Methyl pyrrolidone	199	93	396	202	IIIA
Benzyl alcohol	213	101	401	205	IIIB
Caproic acid	215	102	400	204	IIIB
Ethylene glycol	232	111	388	198	IIIB
3-Ethylhexyl lactate	235	113	475	246	IIIB
Phenyl ether	239	115	498	258	IIIB
Stearic acid	385	196	726	386	IIIB

11.3.1.2 Combustion

The combination of conditions that can allow an explosion are known as flammability. *Flammability is not the same as flash point.* But both are measurements about combustion. Both are important to users of solvents in cleaning operations.

Three factors must be present in a situation for combustion to occur. All three are present in both flash point and flammability testing. All three can be present in cleaning operations.

- There must be a *fuel* present. This is the vaporized solvent, containing at least hydrogen and carbon atoms.

- There must be a source of *oxygen*. This is usually air. But it could be pure oxygen in a closed system. This is simply why hydrocarbon solvents are not used to clean tubing which conveys NASA's liquid oxygen rocket fuel or the U.S. Navy's deep sea breathing gas mixtures.
- There must be a *source of ignition*. This is usually a spark produced unintentionally by mechanical means. But it can be an electrical discharge produced intentionally, or an unexpected hot surface (see below).

Both flammability and flash point relate to combustion of hydrocarbon-based compounds (containing at least hydrogen and carbon atoms) with oxygen in the air. That is something users of solvents want to avoid in all operations!

11.3.1.2.1 Flammability Limits

A vapor mixture is flammable when there are the correct proportions of fuel vapor and oxygen present. If there is too much or too little of either fuel or oxygen, a mixture will not be flammable. There must be a minimum concentration of fuel molecules for a fuel vapor–air mixture to be flammable.

The concept of too little oxygen (lean mixture) or too much oxygen (rich mixture) is crucial. An explosion will only happen, given an ignition source, if the oxygen/fuel ratio is within certain limits. For simplicity, those ratio limits are expressed in terms of fuel (solvent) concentration.

The minimum solvent (fuel) concentration in air is called the lower explosive limit, or LEL. The maximum concentration in air is called the upper explosive limit, or UEL. A match held over a solvent of low volatility may not produce a flame because enough solvent has not vaporized for the LEL to be exceeded. A match dropped in a liquid solvent probably will not produce a flame because the solvent concentration in air is well above the UEL. A match exposed to a solvent air mixture will produce a flame when the concentration in air is above the LEL and below the UEL.

Any covered vessel containing liquid solvent produces a solvent (fuel) mixture in air when the liquid evaporates. The vessel can be a railroad tank car, a paint can, a storage tank, a vapor degreaser, transfer piping, or a cold cleaning tank. The location of this potentially flammable mixture in the vessel is called the headspace, or less commonly, ullage.

11.3.1.2.2 Flammability Limits vs. Flash Point

[Table 11.14](#) compares these two measures of potential for ignition. The reason for two measures is that two different circumstances can occur.

11.3.1.2.3 Flammability Test Equipment

The ASTM E681⁴² apparatus includes a spherical glass flask of a specific volume. The flask has a stirrer for mixing the materials. The flask also has an ignition source, which can be a match, spark, or hot wire. The top of the flask is sealed with a rubber stopper equipped with inlet tubes for air and solvent (fuel).

TABLE 11.14 Two Different Measures of Ignition Risk

Item	Flash Point	Flammability Limit
Situation under study	Heating of a solvent in an open or closed container	Storage or use with vaporized solvent in a closed container
Variables held constant in test	Strength of ignition source	Temperature (usually 25 °C), strength of ignition source
Variables in test	Temperature (composition is an implied variable as increasing temperature increases solvent vapor pressure, and effectively solvent concentration)	Composition
Test fixture volume	Small cup (~ few cubic centimeters of liquid)	Large spherical vessel (~5 l of vapor)
Detection method	Visual recognition of flame in manual testers; temperature or pressure rise in automatic testers	Visual recognition of flame
ASTM standards	D1310-00 (open) ³⁹ or D 56-05 (Closed) ⁴⁰ for Taglibue tester	E 681-01 ⁴²
Test outcome	Temperature at which a flame is first produced (a stable flame front does not have to be produced)	Volumetric composition at which a stable flame front is produced at temperature
Reason for testing	Testing by solvent manufacturers to characterize product vs. shipping, use, and disposal regulations before it is released	Testing on site by users of headspace composition analyzers vs. LEL adjusted with safety factor
Use of outcome	Based on regulations, choice of shipping and packaging methods; design and specification of cleaning equipment and installation details; waste disposal method	Modification of site equipment and procedures to avoid unsafe condition which could produce explosion

The flask is enclosed in an insulated chamber. A magnetic stirrer drive is connected to the flask.

For each test, the vessel is evacuated and precise amounts of test gases, measured by partial pressures, are added. The inlet tubes are closed and the ignition source is triggered. The upward and outward propagation of the flame away from the ignition source is noted, and the concentration of the flammable

component varied between trials until the composition which will just *sustain propagation* is determined.⁴³

The lower flammability limit (LEL) is determined by finding the lowest concentration of fuel vapor that will result in flame propagation for a given spark. As the spark is made weaker, the concentration of fuel vapor has to be increased for the mixture to remain flammable. The strength of a spark igniter is measured in terms of the stored electrical energy (J) used to create the electrical discharge. The lower flammability limit is typically measured using a spark of energy 10–100 J. This is comparable to the arc created by a short circuit in household wiring.

The same procedure is followed to determine the upper flammability limit (UEL).

11.3.1.2.4 Summary of Flammability Limits

Many users believe, and suppliers imply, that data about flammability limits have something to do with flash point tests. This implies that the two have some relationship. They do not.

Flammability limits are measured with a different sample (vapor vs. liquid), a different apparatus (a large enclosed spherical vessel holding ~5 liters of vapor vs a small open or closed cup), at a different temperature (usually 25 °C vs the flash point), and with a different procedure (ASTM E681-01⁴² vs. ASTM D93-02a⁴⁴).

11.3.1.3 Static Discharge

Static electricity is the electricity trapped on the surface of a nonconductive body. Electricity on a conducting body that is in contact only with nonconductors is also prevented from escaping and is therefore neither mobile nor “static.”

Where liquids flow through a pipe, static electricity is generated. Discharge of static electricity is a release of energy. Sufficient static electricity can act as a spark and cause ignition of a cleaning solvent.

The pipe network can be just the shipping or storage container. The conductive properties of the liquid and the pipe network system affect the generation (charging) process. When liquids flow through closed metal pipes, static electricity is not a hazard. This is because the liquid surface is already contacting the metal pipe.

It may become a hazard, however, when liquids are pumped into tanks. In some cases, it is necessary to restrict flow rates to control static generation. Charges produced in the liquid during pumping can accumulate on the surface of the liquid and cause sparking between the liquid surface and the tank or a projection in the tank.

Filters in pipelines greatly increase the generation of static electricity. In one aircraft fueling test, it was reported that the charge development was 10–200 times more with a filter than without a filter.⁴⁵

There is a basic requirement for the grounding (earthing) of process equipment to prevent ignition of flammable vapors by static discharge.

The dielectric constant (DC) of a solvent is a measure of the relative effectiveness of that liquid as an electrical insulator. A good electrical insulator will allow static charge to build in a piping network, and not be dissipated. This is what we *do not* want to happen, to avoid ignition.

A solvent with a low DC is more of a safety hazard than one with a high DC. Static charge dissipation can produce a spark which can ignite an air/solvent mixture. The perfect electrical insulator is a vacuum, which has a DC of 1.00000. By comparison, air has a DC of 1.00059, almost the same as a vacuum, and water has a DC value of 78.2. For a liquid, water is a poor insulator. A solvent with a DC of 10 will allow dissipation of 10 times the amount of static electricity as will a solvent with a DC of 1.

A dielectric meter measures the relative DC of a solvent by measuring the difference in capacitance of the probe between a standard (usually cyclohexane, with true DC value of 2.025) and the solvent sample. Recall that cyclohexane is a commonly used cleaning solvent. As an extremely poor dissipater of static electricity, its use raises significant safety concerns. This is true whether the cyclohexane is used alone or azeotroped with another chemical such as isopropyl alcohol.

The tendency to discharge static electricity can be compared between various solvents with the DC data in [Table 11.15](#). Higher DC values mean static charge is more easily discharged as a spark and represent a solvent that is safer to use.

11.3.1.4 Procedures Recommended to Avoid Fires

The basic approach is to control two of the three factors necessary to start a fire. The factors are: oxygen, fuel in a ratio between the lower explosive limit (LEL) and the upper explosive limit (UEL), and a spark. Perfect control of only one of the three factors will prevent ignition or explosion. Control of a second factor is strongly recommended for security—because perfection in the control of anything is seldom achieved, and because a mistake with a low DC solvent such as cyclohexane or acetone is equivalent to a fire.

The methodology of preventive fire (or explosion) protection comprises the reliable exclusion of at least one of the requirements necessary for the development of an explosion.

Some manufacturers of equipment, or operators of facilities, may take a less conservative approach than the following. Control of two factors is more expensive than control of one. This author cannot recommend that approach.

The two controlled factors are oxygen (through air exposure) and spark (through grounding of static electricity).

- *Air exposure.* There are two general approaches:
 1. No air is to be allowed in any parts of systems where cyclohexane (or azeotropes) exist. Nitrogen is the normal atmospheric gas. A measurement of oxygen content will prove the nitrogen delivery system is working.

TABLE 11.15 Dielectric Constant Data		
Solvent	Temp (°F)	DC
Acetone	77	1.4
<i>n</i> -Hexane	68	1.9
Heptane	68	1.9
Methylhexane	68	1.9
Cyclohexane	68	2.0
Cyclopentane	68	2.0
Toluene	77	2.0
Dichloroethane	77	2.0
Dichloromethane (Meth)	68	2.1
Dichloroethylene	68	2.2
Limonene	68	2.3
Xylene	68	2.4
Tetrachloroethylene (Perc)	70	2.5
Glycerol	77	2.5
Methanol	77	2.6
Freon 113	70	2.6
Isopropyl alcohol	75	4.2
Ethyl ether	68	4.3
Ethanol	77	4.3
Chloroform	68	4.8
Acetic acid	68	6.2
Methyl acetate	77	6.7
Propyl alcohol	77	6.7
Ethylene glycol	68	7.0
Cyclohexene	68	8.3
Trichlorethylene (TCE)	68	8.5
Trichloroethane (TCA)	68	8.5

2. Process control via continuous measurement of %LEL (in air). Alarms are set for levels such as 25% LEL (in N₂). The alarms produce positive action such as shutdown or flooding with nitrogen.
- *Grounding of static electricity.* This is a discipline with multiple details. A full textbook is necessary to cover all details. Designers of well-grounded facilities will be intimately familiar with OSHA regulations, 1910.107 (spray finishing using flammable and combustible materials), OSHA 1926.404(b)(1) (ground fault interrupters),^{46,47} the NFPA standard,⁴⁸ Process Safety,⁴⁹ and/or site design standards.

The reason this author cannot recommend use of just one of the above factors is security. This author is not willing to stake the avoidance of a fire on the integrity of a ground connection, the continuity of an explosimeter, or the position of a N₂ supply value.

11.3.1.5 Body Contact

Body contact is the second major hazard associated with the use of solvents (and aqueous cleaners) in cleaning or other operations.

The contact hazards of a solvent have no direct effect on its capability as a cleaning solvent. This is because the human body is much less able to retard effects of hazardous chemicals than are metals, plastics, glasses, or other substrates which are cleaned with solvents.

Contact hazards, however, have a dominant indirect effect on selection of the cleaning process, cleaning equipment, and cleaning procedures; as well as packaging, transportation, selection, and disposal of the cleaning solvent. In other words, in 1995, when corporate exposure limits for *n*-propyl bromide were between 100 and 200 ppm, considerable cold cleaning work was planned; but in 2007, when corporate exposure limits for *n*-propyl bromide were between 10 and 25 ppm, only cleaning work in enclosed machines was planned.

This subsection is about what can happen if proper precautions are not taken to avoid human contact with solvents. This is not a textbook about internal or external human medicine; a physician should be consulted when and as directed by the MSDS (Material Safety Data Sheet) for the solvent being used. This subsection describes the general contact hazards associated with the use of solvents, and how to prevent them.

11.3.1.5.1 Routes of Entry

Simply put, body contact refers to the three chief routes by which a solvent can enter a human body. Two are inadvertent: inhalation and skin contact, the third is usually intentional or inadvertent ingestion. In the workplace the greatest risk is of skin damage, followed by skin absorption, inhalation, and ingestion of chemicals.

Some chemicals, such as strong acids and alkalis (e.g. chromic acid, sulfuric acid, nitric acid, and sodium hydroxide), produce damage within a very short period of contact. In general, cleaning solvents require prolonged, repeated contact before an effect is seen (e.g. liver damage and cancer by inhaled carbon tetrachloride, leukemia by inhaled benzene, allergic contact dermatitis from toluene, and other chemicals). The effect on the user depends on the toxicity of the chemical, exposure time, amount, and individual susceptibility. Remember that the effects of long-term exposure to many chemicals is unknown.

Human bodies were not designed to protect themselves from hazardous chemicals. They were designed to protect themselves from pathogens (infecting agents).

Finally, not all solvents are hazardous to human bodies. Acetone is used to clean some soils from parts and polish from fingernails. Isopropyl alcohol is used to remove some ionic residues from electronic parts and to disinfect skin. D-Limonene is diluted in water to clean petroleum-based grease from automotive transmission parts and vegetable-based grease from kitchen appliances. And water is the most commonly used of all solvents. But without any information, the safest practice is to assume that all solvents are hazardous to human bodies.

11.3.1.5.2 Inhalation

Odor, a property of nearly all solvents, is the first line of defense for humans. In general, humans can smell the presence of solvent at levels well below when the solvent can start to produce harm.

The cardinal rule is that if you can smell a solvent, use protective equipment to prevent yourself from being exposed to the solvent. Unfortunately, the cardinal rule is not all-inclusive. There are a few exceptions. A significant one is formaldehyde. Formaldehyde is a potent irritant, skin sensitizer, and carcinogen. The current permissible exposure limit (PEL) for formaldehyde is 0.75 ppm and the odor threshold for most people is 1 ppm. Under no circumstances should formaldehyde be used as a cleaning solvent.

11.3.1.5.2.1 Effects of Solvent Inhalation Irritation of nasal passages may be the second line of defense from potential damage of inhaled solvent vapors. Here the damage has already started, depending upon the nature of the solvent and the affected person. In a recognized paradigm, irritation (or other toxicity) generally occurs at a concentration somewhat higher (about 3-10 times higher) than the concentration at which odor is first detected (odor threshold).

Persons perceiving nasal irritation should remove themselves from the environment where they were affected. In this way additional tissue damage may be prevented. Nasal irritation should produce the same response as should detection of odor: removal from the solvent-laden environment. Nasal irritation may

also be a harbinger of collateral damage. Solvent vapors which affect nasal passages often affect the tissue of the eye.

Irritation means that personal protective devices were not used as recommended or were ineffective. Headache, dizziness, and nausea resulting from inhalation may follow. If the condition persists, medical attention should be sought.

11.3.1.5.3 Ingestion

Exposure through ingestion can occur by accident or by consuming contaminated food and drinks or by eating food with contaminated hands. People have unwittingly drunk solvents which have been kept in old, unlabeled drinks containers.

The first line of defense is the taste of the solvent. That should cause one who has accidentally or intentionally ingested some solvent to take notice and raise concern. Vomiting may be induced to expel most solvents. In such cases dilution with water or milk may assist. Anyone who swallows a toxic substance should immediately seek medical attention. Information on the MSDS or container label will prove invaluable.

11.3.1.5.3.1 Effects of Solvent Ingestion Fortunately this contact is less frequent than the other two mechanisms of body contact. Further, the volume of ingested solvent may be low. But the effect may be more lethal. The likelihood of becoming sick from chemicals is increased as the amount of exposure increases. This is determined by the length of time and the amount of material to which someone is exposed—one can drink more than one can inhale.

11.3.1.6 Skin Contact

The human body possesses a number of physical and chemical barriers that prevent entry of pathogens or hazardous chemicals. Of these, perhaps the most important physical barrier is the skin. The skin consists of two distinct layers, a relatively thin outer epidermis and a thicker layer, the dermis. The epidermis consists of several layers of tightly packed epithelial cells that are dead and filled with a waterproof protein called keratin. Therefore, it acts as a physical barrier against entry of hazardous chemicals into the body. The dermis contains a gland called the sebaceous gland that produces an oily secretion called sebum. Sebum consists of a number of organic acids that maintain the pH of the skin between 3 and 5. Therefore, intact skin not only prevents entry of pathogens or hazardous chemicals but also inhibits the growth of most pathogenic bacteria due to its low pH.

The skin, however, does not cover the entire surface of the human body. Conjunctiva of the eye, alimentary, respiratory, and urinogenital tracts are

not covered by dry, protective skin but by mucous membranes. Therefore, these locations on the body function as potential entry sites for pathogens or hazardous solvents.

11.3.1.6.1 Effect of Solvents on Human Skin

Skin exposure to organic solvents can cause several problems. The three major ones are: irritant contact dermatitis, allergic contact dermatitis, and scleroderma. The most common problem is the former—irritant contact dermatitis. Industrial workers exposed to organic solvents are notable for demonstrating its effects.

Organic solvents cause skin irritant contact dermatitis in two ways. In “defatting,” the solvent dissolves (cleans) lipids, which are fats and oils, from beneath the outer surface (the epidermis) of skin. This is easily seen as whitening after the skin is rubbed with some solvents. This damage is reversible.

The second is simple: skin irritation. Skin irritation is visualized as erythema (abnormal redness of the skin due to capillary congestion) and edema (abnormal infiltration and excess accumulation of serous fluid in connective tissue), the result of a local inflammatory process. Generally, this damage is also reversible.

Users of cleaning solvents must know whether skin contact with a specific solvent will produce irritant contact dermatitis. General studies have shown that this could be predicted before human exposure via laboratory experiments with animals versus epidemiological studies about past human exposure. The animals were guinea pigs, rats, and rabbits. Often the inside skin of the animal ear is exposed to solvent and observed to see if it was swollen after a certain time interval.

One notable study involved several common cleaning solvents: toluene, *m*-xylene, trichloroethylene, 1,1,1 trichloroethane, *n*-hexane, methyl ethyl ketone, butanol, and acetone. One conclusion of the study would have been expected by students of cleaning science: skin lesions are more severe on skin which is exposed to lipophilic (oil-loving) organic solvents (toluene, *m*-xylene, *n*-hexane, 1,1,1 trichloroethane, and trichloroethylene) and less severe in more water-soluble (methyl ethyl ketone, acetone, and ethanol) solvents.

11.3.1.7 Carcinogenicity

Carcinogens are agents that can cause cancer. Unlike carcinogenic effects, non-carcinogenic effects are believed to have a threshold; that is, a dose below which adverse effects will not occur.

Carcinogenic effects are believed to be caused by cumulative exposure over all levels of dosage. From significant research, toxicologists and doctors believe a small number of molecular events can evoke changes in a single cell that can lead to uncontrolled cellular proliferation. This mechanism for carcinogenesis is referred to as “non-threshold,” since there is theoretically no level of exposure for such a chemical that does not pose a small, but finite, probability of generating a carcinogenic response.

TABLE 11.16 IARC Classification Scheme

Group 1	The agent (mixture) is carcinogenic to humans. The exposure circumstance entails exposures that are carcinogenic to humans
Group 2A	The agent (mixture) is probably carcinogenic to humans. The exposure circumstance entails exposures that are probably carcinogenic to humans
Group 2B	The agent (mixture) is possibly carcinogenic to humans. The exposure circumstance entails exposures that are possibly carcinogenic to humans
Group 3	The agent (mixture or exposure circumstance) is not classifiable as to carcinogenicity in humans
Group 4	The agent (mixture or exposure circumstance) is probably not carcinogenic to humans

A chemical is considered to be a carcinogen if:

- It has been evaluated by the International Agency for Research on Cancer (IARC) and found to be a carcinogen or potential carcinogen (see [Table 11.15](#)); or
- It is listed as a carcinogen or potential carcinogen in the Annual Report on Carcinogens⁵⁰ published by the National Toxicology Program (11th edition); or
- It is regulated by OSHA as a carcinogen.

Tests with animals and epidemiological analysis suggest that certain chemicals are, or might reasonably expect to be, human carcinogens.⁵¹

It should be noted that some manufacturers and some users do not agree with the selection of chemicals on these two lists.

The IARC classification system of risk to humans has four groups, as shown in [Table 11.16](#).

Most classified chemicals are not cleaning solvents because users have been informed of this classification. Those chemicals which are or have been cleaning solvents and are “Reasonably Anticipated to be Human Carcinogens” are: carbon tetrachloride, chloroform, dichloromethane (methylene chloride), 1,2-dichloroethane, tetrachloroethylene (perchloroethylene), tetrafluoroethylene, and trichloroethylene.

11.3.1.8 Protection from Hazards

There are two basic ways users can cope with whatever hazards are involved with cleaning solvents. The first way is to become informed. The second

way is to take, and continue to take, whatever action is required and justified by the available information.

While there are a great variety of useful cleaning solvents, which impose a variety of hazards in use, there are two cardinal rules to protect workers from these hazards:

- Meet the stated exposure limit, or do not use a cleaning solvent which has no exposure limit.
- Do not come into physical contact with the solvent.

11.3.1.8.1 Becoming Informed

Information exists to inform. If not used, it is wasted.

OSHA requires information about chemical hazards to be provided in MSDSs and labels. These sources are intended to be tools by which hazards of using solvents (and all chemicals) can be identified by users, communicated to workers, *and used to protect workers from those hazards*. Unless it is so used, this information has no value.

11.3.1.8.1.1 MSDS The MSDS is intended to be a tool by which hazards in using chemicals can be identified by users and communicated to workers. In practice, the MSDS is more likely to be used as a marketing tool by the solvent manufacturer or distributor.

More than 25 years ago, OSHA recognized a need for users to have a standardized method of learning about the hazards associated with chemicals used by their workers.⁴³ This information would allow users to make an informed choice about what chemicals they used. In theory users could compare the composition of products and chose those products which impose less risk of injury in use.

11.3.1.8.1.2 Required Content The provisions of this regulation⁵² for MSDSs are significant to users of cleaning solvents. They are as follows.

1. While an MSDS is required only for chemicals imposing hazards to workers, in practice all chemicals except air and water have MSDSs. Chemical manufacturers, importers, and employers must have an MSDS for each hazardous chemical which they make, import, or use.⁴³ There is no standard format required for MSDSs, but one is recommended.⁵³ Another reference is American National Standard for Hazardous Industrial Chemicals-Material Safety Data Sheets-Preparation, available from American National Standards Institute (ANSI). The standard number is Z400.1.⁵⁴
2. The definition of hazard is quite general. “Hazardous Chemical” means any chemical whose use produces a physical hazard or a health

hazard (29 CFR-1910.1200c⁴³). For health hazards, the existence of evidence which is statistically significant by standard scientific methods is the enabling factor which establishes a hazard (29 CFR-1910.1200d⁴³).

3. Direction to manufacturers is straightforward (29 CFR-1910.1200d⁴³). It states, “Chemical manufacturers and importers *shall evaluate* chemicals produced in their workplaces or imported by them to determine if they are hazardous ...” and “... *shall identify* and consider the available scientific evidence concerning such hazards.”
4. If there is no scientific evidence about hazards for the chemical product, the hazards are assumed to be that of the product’s components. In this case, components present at less than 1% (weight or volume) may be ignored. However, if the scientific evidence is about carcinogenicity, then only components present at less than 0.1% (weight or volume) may be ignored. MSDSs issued by manufacturers for sale of the chemicals in New Jersey must comply with a more stringent requirement. In fact, the information sheets are called Hazardous Substance Data Sheets (HSDSs). A specific format is required. HSDSs are frequently more thorough and accurate. The most significant difference between MSDSs and HSDSs is that MSDSs are prepared by manufacturers but HSDSs are prepared by professional panels including one physician, one toxicologist, and one industrial hygienist. A total of 2053 HSDSs have been prepared. For more details, see Ref. 55 especially when you are considering use of a new product in New Jersey.

One effect of these definitions is that any substance present at more than 1% or 0.1% (for carcinogens) must be identified in the MSDS. It is only through this provision that users learn the ingredients contained in a chemical product.

11.3.1.8.1.3 Required Use Employers are required to provide information to their employees about the hazardous chemicals to which they are exposed, by means of a hazard communication program, labels and other forms of warning, material safety data sheets, and information and training (29 CFR-1910.120 [b]⁴³). The MSDS is one (required) source of that information. A good source for formulated cleaning products is Ref. 56.

11.3.1.8.1.4 The Problem with MSDSs What this regulation has produced is not what one would expect. It has not produced informed purchasers of chemical products or informed workers who use them. This is because MSDSs do not contain all the information users need to make good decisions, select products, and inform workers of all hazards.

11.3.1.8.1.5 The Situation for Manufacturers Manufacturers have three imperatives: (1) *identify and evaluate* hazards as above, (2) promote their

products through their MSDS, and (3) thwart competitive products as described in their MSDSs.

In general, manufacturers do this by “selective identification or evaluation” (words of the author) of hazards.

Most MSDSs contain incomplete information about the products they describe. The reader can ascertain this for themselves by reading, on the internet, an MSDS and an HSDS for the same product. It is the strong recommendation of this author that HSDSs be used for any evaluation of chemical products.

11.3.1.8.2 Sources of Information

Hazard information comes from more than MSDSs and labels. These sources may not be complete. This is because MSDSs are both communication and marketing tools, and labels do not have multiple pages. Consequently, do not limit your accumulation of information to that provided with the solvent.

Today, there is another method of communication of information—the Internet. Go to your favorite search engine and type something similar to the following phrase:

“n-methyl pyrrolidone” health, safety environment EPA OSHA

The response may be something you needed to know about *n*-methyl pyrrolidone (for example), such as:

- There is a better choice of solvents (less hazardous to humans or to the environment, better performance, or lower cost). This search phrase will undoubtedly uncover alternative solvents or cleaning processes which may be new to your workers.
- Experiences from users such as yourself. You may learn that Solvent X was “... OK. We proved for 2.5 years that it could be used. But it was a stepping stone. As with any product, you need to be looking for something better ...”
- A forthcoming regulation which will affect how you use that solvent.
- How to recycle it so your costs are reduced.

More to the point, encourage your workers to do this search on the Internet as part of the communication process.

An excellent source of information about solvent (chemical) hazards has been developed by NIOSH.⁵⁷ It is superior to any MSDS—and it is free! There is no excuse for anyone doing solvent cleaning work for not having a copy.

11.3.1.8.3 Communication of Information

There are some governmental requirements in the U.S. There is an informal Globally Harmonized System (GHS) for the classification and labeling of

hazardous chemicals that is the goal of an effort by the U.S. and other countries to promote common, consistent criteria for classifying chemicals according to their health, physical and environmental hazards, and to develop compatible labeling, material safety data sheets for workers, and other information based on the resulting classifications.⁵⁸ No cleaning work is exempted from these requirements. Per the Federal Register No. 59:6126-6184⁵⁹:

.... there were still some comments submitted which suggested that certain industrial sectors should be exempted from the rule, or only covered by limited provisions. The majority of these were from representatives of the construction industry, and from distributors of hazardous chemicals. The arguments generally involved the degree of risk encountered in the industry, and the feasibility of the requirements. OSHA has not found the arguments regarding infeasibility to be persuasive, nor is there any justification for lessening the protections afforded employees in the industries in question ...

Here are the basics of what you are required to do (in the U.S.):

- *Communicate* information about chemical hazards. This communication must be *written*. In that way information can be accessible in an emergency.
- *Train* those affected in how to protect oneself from those hazard.
- *Maintain files* of the communicated information. Failure to meet this requirement is an excellent way to obtain a citation from an OSHA inspection!
- *Maintain on-site records* of those communications and training. In addition this author strongly recommends that you:
- *Repeat* the training on an annual basis.

Any method of Hazard Communication (HAZCOM) training can be used. But the language of the worker must be used.

It is not the responsibility of employees to train themselves. The requirements are not satisfied solely by giving employee the data sheets to read. The employer maintains the responsibility to ensure that their employees are adequately trained and equipped with the knowledge and information necessary to conduct their jobs safely.

The written program must reflect what employees are doing in a particular workplace. For example, the written plan must list the chemicals present at the site, indicate who is responsible for the various aspects of the program in that facility and where written materials will be made available to employees. The written program must describe how the requirements for labels and other forms of warning, material safety data sheets, and employee information and training are going to be met in the facility.

You do not have to develop your own written program. A general course has been developed by OSHA's Mine Safety Health Administration (MSHA) for miners.⁶⁰ There is also an on-line, 11-lesson training course.⁶¹ Each lesson covers a different topic related to the HAZCOM rule. The course can be completed in one sitting or one lesson at a time.

This Hazard Communication requirement pre-empts all state or local regulations (in the U.S.).

11.3.1.9 *Taking Action*

The earlier discussion on body contact and on flammability is not meant in any way to dissuade users from use of cleaning solvents versus other technology.

The choice to use solvent cleaning versus aqueous or other cleaning should depend upon the application, not upon fear of the consequences of hazards. These hazards have been, can be, and will be overcome by users taking appropriate precautions. Rather, that information is meant to define the hazard so that the correct precautions can be taken. Solvent cleaning is being done safely in the U.S. A reliable estimate is that there are at least 10,000 solvent cleaning units or machines being used in the U.S.

Body and flammability hazards are the two major types of general hazards presented to users of cleaning solvents. In addition, there are at least two other circumstances which might be called hazards.

11.3.1.10 *Legal or Regulatory Hazards*

Only one chemical has been banned in the entire U.S. based on use: HCFC 141b. The use of other chemicals such as 1,2,1 trichloroethane and CFC-113 is not banned. Rather, in the U.S. it is the manufacture of those chemicals which is banned by the 1990 Clean Air Act (CAA). These bans carry the weight of law in the U.S. because they are regulations from the U.S. EPA supported by an act of Congress (the CAA). Locally or regionally, the use of certain chemicals in certain operations may also be banned for cleaning work.

In any of these cases, a potential or actual user of these solvents is under the hazard of legal sanction. While the author of this chapter recognizes numerous violations of these bans, he cannot recommend their violation in any circumstance. Users or manufacturers of banned cleaning solvents should consider themselves being exposed to a hazard as significant as ignition or ingestion. Here, the technical protection is to seek an alternative or replacement solvent.

11.3.1.11 *Economic Hazards*

If legal sanction is a hazard, then economic loss is one as well. This author is of the opinion that cleaning costs, if the system is well specified, designed, and used, should be almost negligible in the total manufacturing cost.

Users who choose to use a cleaning solvent with abnormal functionality, or about which there is incomplete information about potential hazards, have placed themselves at risk of being exposed to economic hazard. If in doubt, contact the appropriate regulatory agency.

In the decade of the 1990s, this author has seen many firms consciously or unconsciously expose themselves to the hazard of economic damage by mandating a uniquely functional cleaning solvent. Yet only a few percent of those users remained under hazard. The great majority accepted a change in solvent,

process, or specification. Again, technical protection from this hazard is to seek an alternative or replacement solvent, process, or specification.

The two general strategies to containing flammability and body contact are: (1) to avoid at least one (and preferably *two*) of the three factors necessary for ignition and (2) assure (via no violation of exposure limits and minimum physical contact) any body contact is at a level which is not expected to cause significant harm.

OSHA defines their hierarchy of control:

- Engineering controls
- Work practice controls
- Administrative controls
- Personal protective equipment

Recommendation of specific action is beyond the scope of this chapter because of the variety of cleaning solvents which may be used in global applications. But some general advice about protective actions should be strongly considered. Accept the recommendations in the MSDS, and other recommendations from the manufacturer, as starting points. Use your own judgment.

11.4 SOLVENT SELECTION VIA SOLUBILITY PARAMETERS

Powerful solvents do not necessarily bring value to users. If trying to remove fingerprints from a polymer surface, should a powerful solvent be used which would attack the polymer substrate? If using a solvent to remove a soil, a user would want to know that the solvent would dissolve the soil before they bought the solvent and cleaning equipment.

The link connecting the user with the supplier is that the solvent must be chosen by the supplier to (at the very least) dissolve the user's soil. That is what this section is about: using technology to match soils with solvents which will remove them. The technology is solubility parameters. Mainly, three topics will be covered in this section.

- The Kauri Butanol (Kb) test
- The Hildebrand solubility parameter
- Hansen solubility parameters (HSPs)

11.4.1 Background

Today, it is possible to forecast with a high degree of confidence whether or not a particular solvent will dissolve a particular soil from a part. It is not necessary to wet the part. The ability to do this has been evolving for the last half decade or so. Formulators of paints and coatings have been instrumental in these developments. They do not want to remove a resin or polymer as a soil. They want to formulate a product which will apply the resin/polymer to a surface as a coating. The product is the resin/polymer and a suitable solvent which will act as a

carrier to apply the resin/polymer and which can be removed in some drying process. These formulators have valued, developed, and are using solubility parameters.

From the 1930s to the 1970s (roughly), the *Kauri Butanol* (kb) test was the chief method for characterizing the dissolving power of a solvent. Although fundamentally flawed, it was the best available technology.

In 1950s and 1960s and afterward, Hildebrand and coworkers and other researchers pioneered the concept that the molecular forces holding solvent molecules together are the same forces which may engage a soil and possibly dissolve it.^{62,63}

Hansen, from the 1960s through today, enhanced Hildebrand's work by characterizing the total molecular force holding solvents together as the sum of three component forces and separating the Hildebrand solubility parameter into three components.^{64,65}

Solubility parameters are not well known to suppliers of cleaning agents (though they are definitely so to suppliers of components for coatings). Suppliers of cleaning agents generally do not use solubility parameters to match their solvent with the user's soil. This is because suppliers apparently believe solubility parameter technology is too complicated for users to understand, and suppliers apparently feel that providing solubility parameter information about their soils may provide advantage to competitors while only adding cost to themselves.

11.4.2 The Kauri Butanol Test

This parameter (kb) is a measure of solvent strength used only for hydrocarbon solvents; it has no meaning for polar solvents (chiefly, oxygenated solvents).

The test, though common and long-standing, is an indirect one. The kb test does not directly measure the general solvent power of a test solvent. This cannot be done with a single solvent and a single soil.

The kb test is based on the observation that when a non-solvent is added to a solution containing a solvent and a dissolved soil, the soil precipitates from the solution. The amount of soil precipitation is dependent, at a given temperature, upon these two parameters.

- The degree to which the non-solvent is not a solvent for the soil. More soil will be precipitated if the non-solvent has less power to dissolve the soil. Said another way, all the soil would be precipitated if the non-solvent had no solutioning power for the soil.
- The volume of non-solvent added to the solution.

Kauri gum (resin), a natural product, was once a major component of lacquers, varnishes, enamels, and other coatings. Paints based on linseed oil were hardened by the addition of such materials as Kauri gum. Kauri gum is readily soluble in *n*-butyl alcohol and other polar solvents. But it is not readily soluble

TABLE 11.17 Ranking of Solvents by Kb Values

Solvent	Kb Value
PFCs	0 (no power to dissolve Kauri gum)
HFC 43-10mee & HFE 7100	~10
Siloxanes	~15
Typical aliphatic hydrocarbons	Between 25 and 40
CFC-113	32
Soy-derived solvents	55–60
HCFC 141b	56
Typical aromatic hydrocarbons	Above 70
Chlorinated and brominated solvents	Above 100
Alcohols, ketones, dibasic acids, esters, ethers, glycol ethers, etc.	Test has no meaning because Kauri gum is completely soluble in these solvents

in hydrocarbon solvents. So the kb test can measure the degree to which a chosen hydrocarbon solvent can dissolve Kauri gum.

Data in [Table 11.17](#) show how various types of solvents are rated by the kb test.

The kb test compares the solvency of various hydrocarbon solvents for Kauri gum. Suppliers (and some consultants) use kb data to compare the solvent power of various solvents. But these comparisons may be meaningless to users *unless the soil in the application is comparable to Kauri gum (which is almost never so)*. Unfortunately, most soils have little in common with Kauri gum. Few users have it on their parts. The kb test was developed to aid those who formulate coatings, not those who design, market, or use cleaning solvents.

There are similar tests with similar values and limitations. Other empirical solubility scales include the aniline cloud-point (aniline is very soluble in aromatic hydrocarbons, but only slightly soluble in aliphatics), the heptane number (measures the amount of heptane that can be added to a solvent/resin solution), the wax number (measures the amount of a solvent that can be added to a benzene/beeswax solution), and others.⁶⁶

11.4.3 The Hildebrand Solubility Parameter

The transition from solvents and soils to kb to “cohesive energy density” and to multiple solubility parameters is due to Joel Hildebrand and those who have built on his work. Hildebrand’s ideas and the work derived from them allow a useful system of solubility characterization.

Hildebrand's basic idea was that there was an energy transaction within a fluid when a solution occurs. Here, the solvent molecules must overcome intermolecular forces in the solute (soil). Solvent molecules must find their way between and around the solute molecules. And solvent molecules themselves must be separated from each other by the molecules of the solute.

Hildebrand's work has two tenets.

- To overcome intermolecular forces, energy is involved. Hildebrand, and those who built on his ideas, showed that these energy requirements were at a minimum if the solute (soil) and solvent exerted the same forces upon one another. Stated another way, "like dissolves like."
- The energy involved with combining a solute (soil) into a solvent is the same as the energy holding the molecules of the solvent together. And the energy holding molecules together against intermolecular forces is the energy required to separate all the molecules from one another. And that is the heat of vaporization. Every intermolecular force must be overcome to vaporize a solvent. So the best estimate of the energy involved with solution of a solute (soil) into a solvent is the heat of vaporization of the solvent.

Said another way, the same intermolecular attractive forces have to be overcome to vaporize a liquid as to dissolve something in it.

What happens when two liquids are mixed? The molecules of each liquid are physically separated by the molecules of the other liquid, similar to the separations that happen during vaporization. The same intermolecular forces must be overcome in both cases.

The energy transaction is described by the following equation⁶²:

$$C \equiv \frac{\Delta H_{\text{vap}} - RT}{V_m} \quad (11.24)$$

The term ΔH is the heat of vaporization with units of energy per mole. The term V_m is the solvent volume per mol, and C is called the "cohesive energy density" which has the units of pressure (for ideal gases, $PV=RT$). The cohesive energy density of a liquid is a numerical value that indicates the energy of vaporization. It is a direct reflection of the forces holding the molecules of the liquid together.⁶⁷

11.4.3.1 A One-Dimensional Solubility Parameter

In 1936, Hildebrand proposed the square root of the cohesive energy density as a numerical value indicating the solvency behavior of a specific solvent. The term "solubility parameter" was proposed for this value and the quantity represented by delta (δ). Then:

$$\delta \equiv \sqrt{\frac{\Delta H - RT}{V_m}} \quad (11.25)$$

The units of this parameter are (pressure)^{1/2} in megapascals (MPa); or calories^{1/2}/cc^{3/2}; or in honor of its creator, the *Hildebrand*. 1 Hildebrand = 1 MPa = 2.0455 calories^{1/2}/cc^{3/2}.

Note that there is a temperature sensitivity to Hildebrand's solubility parameter. Solubility parameters decline with temperature – the product RT increases very slightly as compared with the declining value of the heat of vaporization.

Hildebrand's δ :

- ✓ Does yield absolute results (it is not a calibrated or relative result),
- ✓ Is not specific to one solvent or soil,
- ✓ Does yield information, when values are compared, about the character of solvents, and
- ✓ Does allow predictions about solubility when compared to values from soils.

Stated another way, a solvent can dissolve a soil when the δ values for the solvent and the soil are the same.

Excellent solvent cleaning results when:

$$\delta_{\text{solvent}} = \delta_{\text{soil}}$$

The difference between δ for a solvent and δ for the soil is a measure of the difficulty of dissolving (cleaning) the soil with that solvent.

Poor solvent cleaning results when:

$$\delta_{\text{solvent}} \langle \langle \delta_{\text{soil}} \quad \text{OR} \quad \delta_{\text{solvent}} \rangle \rangle \delta_{\text{soil}}$$

Users have been taught to choose their cleaning solvent to have the same, or similar, Hildebrand solubility parameter as does their soil.

δ is described as a one-dimensional solubility parameter because it is based on the total of the intermolecular forces within a solvent.

11.4.3.2 Molecular Forces

In practice there are multiple types of forces which act between the molecules of a solvent. These attractive and repulsive forces between molecules are called *van der Waals forces*, first described by Johannes Diderik van der Waals in 1873. These forces do not bind together quantum particles in atoms. They exist between and among molecules of the same substance. Some thought these forces were small gravitational attractions. van der Waals forces are actually due to electromagnetic interactions between molecules. These forces exist because molecules are not uncharged round objects which pack a void and act on the macro scale as a continuum. Molecules of any substance have various shapes – some small and simple (methylene chloride), some large and not simple (glycol ethers).

The envelope containing a single molecule is not uniformly charged (attractive or repulsive). Sections of the boundary surrounding a halogen atom are rich

with electrons. Sections surrounding a hydrogen atom are rich with positive electrostatic force (absence of electrons).

There are three types of intermolecular (not intramolecular) forces:

- *Polar interactions.* These dipole-dipole forces exist on molecules that are slightly charged on each side. A dipole is a pair of magnetic poles, each with opposite charge, separated by a short distance. In water, the oxygen part of the molecule is negatively charged, and the hydrogen side of the molecule is positively charged. This force makes groups of like molecules tend to “stick” together, and makes it difficult for that group to accept (dissolve) another molecule which does not exert those electrostatic charges.
- *Hydrogen bonding forces.* These are a special type of dipole-dipole interaction. A hydrogen bond is a dipole-dipole interaction that occurs between any molecule with a bond between a hydrogen atom and any of oxygen/fluorine/nitrogen atoms. The dipole exists because oxygen, nitrogen, and fluorine are extremely good at attracting electrons and hydrogen is extremely good at losing them. The extremely positive (hydrogen) side of the molecule will orient itself with the extremely negative (oxygen, nitrogen, or fluorine) side of another molecule. Hydrogen bonding forces, an extreme dipole situation, are extremely strong – perhaps 1/10 the strength of a covalent bond.⁶⁸ Hexane, ethers, ketones, HFEs, and HFCs do not have substantial hydrogen bonding forces; alcohols/glycols, water, acids, and ammonia do. There is no OH bond in ketones or ethers.⁶⁸
- *Dispersion forces.* Quantum theory helps explain dispersion forces. This teaches that the components of atoms are not static – are not always in the same place all the time. They move at high speed over very short distances. On average, the envelope containing a single molecule has a constant condition of electrostatic charge. But at *any* moment, *every* portion of that envelope will have a changing electrostatic environment. Forces of short duration – temporary forces – are produced. They are called dispersion forces, and are of considerably less magnitude than are dipole-based forces. The degree of “polarity” that these temporary dipoles confer on a molecule is related to surface area. Larger molecules have a greater number of temporary dipoles and greater intermolecular attractions. Molecules with straight chains have more surface area, and thus greater dispersion forces, than branched-chain molecules of the same molecular weight. These induced attractions are also called London dispersion forces, or induced dipole-induced dipole forces.⁶⁹

11.4.4 Hansen Solubility Parameter

Dr. Charles Hansen writes, “... needless to say, without the work of Hildebrand and Scott and others ... this postulate could never have been made”.⁶⁵ He was referring to the idea that the total energy of vaporization of a solvent consists of several individual components each of which arises because of an intermolecular force. Chiefly, these are the same three forces described above.

Basically, Hansen splits the Hildebrand solubility parameter (HSP) into three component parameters. Each component is associated with an inter-molecular force: dipole-dipole, hydrogen-bonding, and dispersion. HSPs come from equating the total cohesive energy of a solvent to the sum of three cohesive energy terms—each one representing one of the component forces.

$$(\Delta H_{\text{Vap}} - RT) = E_{\text{Polar}} + E_{\text{Hydrogen-Bonding}} + E_{\text{Dispersion}} \quad (11.26)$$

Dividing each term by the molar volume of the solvent, one gets the defining equation:

$$\delta^2 = \delta_{\text{Polar}}^2 + \delta_{\text{Dipole}}^2 + \delta_{\text{Dispersion}}^2 \quad (11.27)$$

This “Pythagorean Theorem” of solubility parameters allows decomposition of the solubility parameter developed by Hildebrand into three components, each one representing one of the component forces.

Hansen’s approach solves a fundamental problem with Hildebrand solubility parameters. Some solvents have equivalent values of Hildebrand solubility parameter. But their molecular structures are not similar. The inter-molecular forces between molecules are not similar. Their solubility performance is not the same. Examples of solvents which have the same Hildebrand solubility parameter but do not have similar structures include chloroprene and di-isobutyl ketone; cyclopropane and dimethyl cellosolve; xylene and ethyl acrylate; and toluene and 1,1-dichloropropane, to name but a few pairs.

It is this problem, with its solution, which makes Hildebrand solubility parameters of immense theoretical but only limited practical value, and which makes HSPs of significant value in predicting solution behavior between soils and solvents.

Hansen’s initial work was trial and error. Today HSPs are determined solely from experiments or calculations using the data one can find for latent heat, dipole moment, group contributions, etc.

11.4.4.1 Solvent Substitution with HSP

11.4.4.1.1 HSP Data

There are two sets of solubility parameters: one for solvents, and one for soils. There is no value in cleaning work of having one set of three solubility parameters. It does not matter per se whether the values of solubility parameter are large or small. Absolute values have no meaning. Stated another way, the concept of a powerful solvent is a false one. A solvent is “powerful” if its solubility parameters are similar in value to those of many soils.

Solubility parameter values have value only when they are compared to values for something else, such as another solvent or a soil. Relative values have meaning. HSPs for some polymeric soils are shown in [Table 11.18](#).⁷⁰

Data for selected cleaning solvents are given in [Table 11.19](#). $\delta_{\text{Hildebrand}}$ is computed from Eq. (11.27).⁶⁵ Note how the value of $\delta_{\text{Hildebrand}}$ is essentially

TABLE 11.18 Hansen Solubility Parameters for Polymeric Soils				
	Commercial Product	$\delta_{\text{Dispersion}}$ Values in MPa ^{1/2}	δ_{Polar} Values in MPa ^{1/2}	$\delta_{\text{Hydrogen-Bonding}}$ Values in MPa ^{1/2}
Cellulose acetate	Cellidore® A, Bayer	18.6	12.7	11
Chlorinated polypropylene	Parlon® P-10, Hercules	20.3	6.3	5.4
Epoxy	Epikote® 1001, Shell	20.4	12	11.5
Isoprene elastomer	Ceriflex® IR305, Shell	16.6	1.4	−0.8
Cellulose nitrate	1/2 sec, H-23, Hagedorn	15.4	14.7	8.8
Polyamide, thermoplastic	Versamid® 930, General Mills	17.4	−1.9	14.9
Poly (isobutylene)	Lutonal® IC-123, BASF	14.5	2.5	4.7
Poly (ethylmethacrylate)	Lucite® 2042, DuPont	17.6	9.7	4
Poly(methyl methacrylate)	Rohm and Haas	18.6	10.5	7.5
Polystyrene	Polystyrene LG, BASF	21.3	5.8	4.3
Poly(vinyl acetate)	Mowilith® 50, Hoechst	20.9	11.3	9.6
Poly(vinyl butyral)	Butvar® B-76, Shawnigan	18.6	4.4	13
Poly(vinyl chloride)	Vilpa® KR, k=50, Montecatini	18.2	7.5	8.3
Saturated polyester	Desmophen® 850,	21.5	14.9	12.3

TABLE 11.19 Hansen and Hildebrand Solubility Parameters for Solvents

	$\delta_{\text{Dispersion}}$ Values in MPa ^{1/2}	δ_{Polar} Values in MPa ^{1/2}	$\delta_{\text{Hydrogen-}}$ Bonding Values in MPa ^{1/2}	$\delta_{\text{Hildebrand}}$ Values in MPa ^{1/2}
Acetone	15.5	10.4	7.0	19.9
Aromatic 150	18.2	1.0	3.1	18.5
Benzene	18.4	0.0	2.0	18.5
<i>t</i> -Butyl alcohol	15.2	5.1	14.7	21.8
Cyclohexane	16.8	0.0	0.2	16.8
Diethylene glycol butyl ether acetate	16.0	4.1	8.2	18.4
Diethylene glycol monoethyl ether	16.1	9.2	12.2	22.2
Dipropyl ketone	15.8	5.7	4.9	17.5
Ethanol	15.8	8.8	19.4	26.5
Ethyl lactate	16.0	7.6	12.5	21.7
Ethylene glycol diethyl ether (Ethyl glyme)	15.4	5.4	5.2	17.1
Ethylene glycol monoethyl ether (EE)	16.2	9.2	14.3	23.5
Exxate 1000	14.9	5.7	3.1	16.3
Hexane	14.9	0.0	0.0	14.9
Isopar L	14.9	0.0	0.0	14.9
D-Limonene	16.6	0.6	0.0	16.6
Methanol	15.1	12.3	22.3	29.6
Methyl acetate	15.5	7.2	7.6	18.7
Methyl ethyl etone	16.0	9.0	5.1	19.1
Methylene chloride	18.2	6.3	6.1	20.2
Naphtha high-flash	17.9	0.7	1.8	18.0
Pine oil	15.6	3.0	9.8	18.7
<i>n</i> -Propyl bromide	16.0	5.8	4.2	17.5
Propylene glycol monoethyl ether (PGEE, PE)	15.7	6.5	10.5	20.0

Continued

TABLE 11.19 Hansen and Hildebrand Solubility Parameters for Solvents—
Cont'd

	$\delta_{\text{Dispersion}}$ Values in MPa ^{1/2}	δ_{Polar} Values in MPa ^{1/2}	$\delta_{\text{Hydrogen-}}$ Bonding Values in MPa ^{1/2}	$\delta_{\text{Hildebrand}}$ Values in MPa ^{1/2}
Propylene glycol monomethyl ether (PM, PGME, MPM)	15.6	6.3	11.6	20.4
Soygold 1000	16.2	4.9	5.9	17.9
Tetrachloroethylene (PERC)	18.3	5.7	0.0	19.2
1,1,1-Trichloroethane (TCA)	16.8	4.3	2.0	17.5
Trichloroethylene (TCE)	18.0	3.1	5.3	19.0
Water	15.5	16.0	42.3	22.7
Xylene	17.6	1.0	3.1	17.9

the same for ethyl lactate and *t*-butyl alcohol, while their structures essentially are not the same.

Additional data for more than 200 cleaning solvents will be published in a forthcoming book by the author.¹ More than 60 properties of each solvent will be included in that book.

There are numerous modifications of Hansen’s work. And there are other solubility parameter systems. A unique one will be to devise a single characteristic will be able to separate/predict toxic and nontoxic solvents. An interesting and lengthy paper on this topic can be found in Ref. 71. In practice, the HSPs are the most common ones which are used in critical or metal cleaning applications. The Hansen approach is also claimed by Hoy in a separate publication.⁷²

11.4.5 Solvent Substitution

11.4.5.1 Multiple Components

Nearly all soils contain more than one component. Motor and lubricating oils, coolants, food and medical debris, and solder residue are familiar soils and all contain multiple components. A stereotypical soil may have three components—two active ingredients and a carrier fluid.

A poorly chosen solvent may be successful for some soil components, but will not dissolve all components. The properly chosen solvent will dissolve each and all components. This fact makes solvent selection, by any method, a compromise. A useful solvent must dissolve each component. The HSP of each component must be identified by estimation to other solvents or reference to other work.

Information in the previous tables reminds us of the lesson that soils have multiple components. It is possible that there may be no single solvent which is the best choice for each component in a soil. One can usually identify a single best or better choice for formulated soils and products. This is because there is usually some carrier fluid which is the base component for the mixture which is the commercial soil material. But tramp (unexpectedly made by the user in their process) soils normally require a compromise in selection of solvents to produce an overall result where all components are effectively solubilized in a cleaning operation.

The task of solvent substitution (or choice) is not as difficult as it may seem. After all, individual components of a commercial material which becomes a soil have sufficient affinity to be combined into a single, stable product. This means their intermolecular forces must be similar.

11.4.5.2 HSP Data and Calculations

The best source of solubility parameter data is Hansen's handbook on solubility parameters.⁶⁵ These data are used in the following equation for solvent selection. Equation (11.28) is derived by subtracting Eq. (11.27) with HSP data for solvents from Eq. (11.27) with HSP data for soils.

$$R_A^2 \text{Solution} = (\delta_{\text{Polar for solvent}} - \delta_{\text{Polar for soil}})^2 + (\delta_{\text{H}_2\text{-bonding for solvent}} - \delta_{\text{H}_2\text{-bonding for soil}})^2 + 4(\delta_{\text{Dispersion for solvent}} - \delta_{\text{Dispersion for soil}})^2 \quad (11.28)$$

R_A is the distance in solubility parameter space between the soil and the solvent. The aim is for these values to be as close as possible together ($R_A \text{ Solution} \approx 0^{73}$). The independent variable in Eq. (11.28) is the choice of solvent. The dependent variable in Eq. (11.28) is how well the soil is dissolved—how close $R_A \text{ Solution}$ approaches zero.

Finding the number 4 in the last term of Eq. (11.28) may be surprising. There is considerable discussion in the literature about the need for it. It has been found empirically useful over several decades of practical experience. There are two justifications for it: (1) it converts three-dimensional spheroidal plots of $\delta_{\text{Dispersion}}$, $\delta_{\text{H}_2\text{-Bonding}}$, and δ_{Polar} into spherical ones; (2) theoretical considerations associated with the Prigogine Theory of Corresponding States.⁶⁵

An example calculation in Table 11.20 shows that these evaluations are best done with a spreadsheet. Four soil components are considered in Table 11.20.

TABLE 11.20 Example of Details for Solvent Selection with HSP

Type of Soil	$\delta_{\text{Dispersion}}$	δ_{Polar}	$\delta_{\text{Hydrogen-Bonding}}$	Solvent	$\delta_{\text{Dispersion}}$	δ_{Polar}	$\delta_{\text{Hydrogen-Bonding}}$	Dispersion Term	Polar Term	H ₂ Bond Term	RA ²
Vegetable oil (olive)	15.9	1.2	5.4	Acetone	15.5	10.4	7	$4 \times [(15.5-15.9)^2]$	$(10.4-1.2)^2$	$(5.4-7)^2$	87.8
				Ethanol	15.8	8.8	19.4	$4 \times [(15.8-15.9)^2]$	$(8.8-1.2)^2$	$(5.4-19.4)^2$	456
				Isopar L	14.9	0	0	$4 \times [(14.9-15.9)^2]$	$(0.0-1.2)^2$	$(5.4-0)^2$	34.4
				Xylene	17.6	1	3.1	$4 \times [(17.6-15.9)^2]$	$(1.0-1.2)^2$	$(5.4-3.1)^2$	16.9
Motor oil SAE 20 W	14.7	0.4	0.4	<i>t</i> -Butyl alcohol	15.2	5.1	14.7	$4 \times [(15.2-14.7)^2]$	$(5.1-0.4)^2$	$(11.1-0.4)^2$	226
				Hexane	14.9	0	0	$4 \times [(14.9-14.7)^2]$	$(0.0-0.4)^2$	$(11.1-0.4)^2$	0.36
				Propylene glycol monomethyl ether (PM, PGME, MPM)	15.6	6.3	11.6	$4 \times [(15.6-14.7)^2]$	$(6.3-0.4)^2$	$(11.6-0.41)^2$	161
Methyl oleate (a surfactant)				Methanol	15.1	12.3	22.3	$4 \times [(12.8-14.7)^2]$	$(6.2-0.4)^2$	$(6.4-0.4)^2$	621
	12.8	6.2	6.4	D-limonene	16.6	0.6	0	$4 \times [(12.8-15.0)^2]$	$(6.2-0.0)^2$	$(6.4-0.0)^2$	87
				TCE	18.0	3.1	5.3	$4 \times [(12.8-15.0)^2]$	$(6.2-0.0)^2$	$(6.4-5.3)^2$	39.4
				Ethyl lactate	16.0	7.6	12.5	$4 \times [(12.8-15.0)^2]$	$(6.2-0.0)^2$	$(6.4-0)^2$	51
				Water	17.6	1.0	3.1	$4 \times [(12.8-15.0)^2]$	$(6.2-0.0)^2$	$(6.4-3.1)^2$	1400

Phosphate hydraulic fluid (Ester)	14.5	10.4	4.5	Ethylene glycol monoethyl ether (EE)	16.2	9.2	14.3	$4 \times [(14.5-16.2)2]$	(10.4-23.7)	(4.5-14.3)2	100
				Cyclohexane	16.8	0	0.2	$4 \times [(14.5-16.8)2]$	(10.4-23.7) 2	(4.5-0.2)	132
				Methylene chloride	18.2	6.3	6.1	$4 \times [(14.5-18.2)2]$	(10.4-23.7) 2	(4.5-6.1)2	33
				<i>n</i> -Propyl bromide	16.0	5.8	4.2	$4 \times [(14.5-16.0)2]$	(10.4-23.7) 2	(4.5-4.2)2	23

Information in [Table 11.20](#) illustrates one reason why acetone is not used to clean up kitchen waste, alcohols are not used to clean up waste from automotive repairs, and why use of hexane to clean up motor oil is an example of the adage (or Hildebrand’s tenet) that “like dissolves like.”

Overall results for four synthetic soils and their several soil components are given in [Table 11.21](#). The soils are: a food residue, a simple lubricant mixture, a polymeric residue, and a medical residue. Note that these synthetic soils are not commonly produced and are created only as examples. Only the top three (lowest values of R_{A^2}) solvent choices are included in [Table 11.21](#) for each soil component.

Users do not normally consider enough choices. There are hundreds of solvents which could provide useful cleaning service, especially if used in VOC

TABLE 11.21 Solvents Chosen with HSP for Specific Soils			
Soil to Be Cleaned	Soil Component	Solvent Evaluated	R_A Solution for Combination
Food residue	Vegetable oil (olive)	Soygold 1000	14.31
		Diethylene glycol butyl ether acetate	16.29
		Xylene	16.89
	Lard	Trichloroethylene (TCE)	1.46
		Xylene	4.38
		Aromatic 150	5.41
	Packaging material	Propylene glycol monomethyl ether (PM, PGME, MPM)	2.85
		Propylene glycol monoethyl ether (PGEE, PE)	3.28
		Ethyl lactate	7.05
Simple lubricant	Paraffinic oil	Isopar L	0.02
		Hexane	0.04
		D-Limonene	10.22
	Linseed oil	Diethyl ether	2.68
		Mineral spirits	2.69
		Diethylene glycol dibutyl ether	5.64

Continued

TABLE 11.21 Solvents Chosen with HSP for Specific Soils—Cont'd

Soil to Be Cleaned	Soil Component	Solvent Evaluated	R_A Solution for Combination
	Silicone DC 23 oil	Hexyl acetate	12.38
		Diethylene glycol butyl ether acetate	17.13
		Methyl acrylate	18.09
Polymeric residue	Polystyrene LG	Ortho-bromo toluene	9.45
		Ethyl cinnamate	21.11
		Napthalene	26.11
	Coal tar pitch	Benzoic acid	1.42
		DBE-4	11.62
		Diethyl ketone	26.06
	PVOH	Ethylene glycol monomethyl ether (EM, methyl cellosolve)	3.24
		Ethanol	3.44
		Diethanolamine	13.52
Medical residue	Fingerprints	Ethylene glycol monoethyl ether (EE)	49.29
		Methanol	56.62
		Diethylene glycol monoethyl ether	60.72
	Blood serum	Ethanol	538.52
		Ethylene glycol monoethyl ether (EE)	644.75
		2-Propanol (IPA)	697.24
	<i>Lignin</i>	Ethylene glycol monoethyl ether (EE)	160.73
		Diethylene glycol monoethyl ether	180.66
		Ethanol	183.18

TABLE 11.22 HSP of Familiar Solvents and Soils			
Well-Known Solvents	$\delta_{\text{Dispersion}}$	δ_{Polar}	$\delta_{\text{H2-Bonding}}$
1,1,1 Trichloroethane (TCA)	16.8	4.3	2.0
Trichloroethylene (TCE)	18.0	3.1	5.3
CFC-113	14.7	1.6	0
Water (with organic liquids)	19.5	17.8	17.6
2-Propanol (IPA or isopropanol)	15.8	6.1	16.4
Common soil components			
Oils (vegetable, olive)	15.9	1.2	5.4
Oils (paraffinic)	15.0	0	0
Esters	15.9	1.2	5.4
Glycols or alcohols	16.8	9.4	23.3

exempt applications (cold cleaning) in Europe. HSP data allow efficient screening of a nearly unlimited number of solvents in a spreadsheet application.

A challenge to identify the cleaning solvent most well known, or used, would produce at least five solvents. They are different. None of them is a “universal solvent” for some common components of soils.

HSP values for five well-known solvents and four general components of industrial soils are shown in [Table 11.22](#).

HSP information in [Table 11.22](#) suggests that:

- There are better solvent choices than those in [Table 11.22](#) for cleaning paraffinic oils, though CFC-113 might be a better choice than expected (if its manufacture were not banned by the Montreal Protocol).
- TCA and TCE differ by TCE being expected to have to more affinity for esters.
- Water is a unique molecule.

In one sense, technology often only lets users quantify what is already, or should be, known before its use. The ancient principle “like dissolves like” (also stated as “like seeks like,” referring to self-association, or “take it off the way it went on,” referring to the point that there is some common solutioning material in a formulated product) has long been used for solvent selection. HSPs quantify this principle and let users efficiently implement this principle over a broad and deep population of solvents. After all, the better solvents (lowest values of $R_{\text{A Solution}}$ identified in [Table 11.21](#)) for olive oil as a soil component are close relatives of olive oil.

11.5 CHOOSING CLEANING SOLVENTS AND CLEANING PROCESSES

There are so many types of solvents that it can be difficult to discriminate among them. Simple solvents such as hexane, more complex solvents such as *n*-methyl pyrrolidone, flammable solvents such as acetone, carcinogenic solvents such as benzene, ozone-depleting solvents such as carbon tetrachloride, expensive solvents such as HCFC 43-10 mee, and low-cost common solvents such as water, have all been used as cleaning solvents.

One of the strengths of solvent cleaning is the variety of types of solvents which can be used in various solvent cleaning processes.

Many users choose their cleaning solvent because it is recommended by a supplier who treats them well, or because its price is low, or because someone else is using it. That choice may be acceptable if, by coincidence, that solvent is matched to the user's soil.

Both the cleaning process and the cleaning solvent can be modified to compensate for their individual deficiencies.

- A solvent which is flammable, but easily dissolves the soil and rinses the part structure and is rapidly evaporated, can be safely managed in a process with complete grounding and inerting and safety interlocks. Firms have sold hundreds of such solvents for many years. If acetone (highly flammable) is the right solvent, get the right process, be certain it is right, and use both.
- A solvent with a high price or a solvent which is a VOC, but easily dissolves the soil and rinses the part structure and is rapidly evaporated, can be contained in a vacuum-based cleaning machine in which there is no interface between the atmosphere and the solvent.
- A solvent with an extremely low surface tension, such as dense-phase liquid or supercritical CO₂, can be used to penetrate complicated part structures. But its dissolving power must be matched to the soil and it is a gas at atmospheric conditions. Consequently, firms select and add cosolvents or detergents which are soluble in liquid CO₂ to improve solvency. The parts are loaded into racks and inserted into a sealed vessel which contains the dense-phase CO₂.
- A simple, low-priced solvent such as a paraffinic hydrocarbon is an excellent solvent for most oils and greases. It is commonly used in a small, open tank ("sink-on-a-drum") to clean brake, engine, or transmission parts prior to repair. Fortunately, drying rate is not an issue since this solvent evaporates only slightly faster than does mercury.

The theme connecting these four (of many) examples is the broad variety of useful solvent cleaning processes. A great variety of solvent types can do cleaning jobs when they are combined with the appropriate equipment.

The combination of the solvent, the equipment, and the method by which both are used is the cleaning process.

One of the major changes in industrial cleaning which occurred during the phase out of ozone-depleting compounds was that cleaning work had been done by a solvent and is now being done in a process of which a solvent is a component. For example, many users thought that when they switched from cleaning with an ozone-depleting solvent to aqueous cleaning that the process would be the same. That was not so!

11.5.1 Choices of Solvents

11.5.1.1 Chemical Structure and Atomic Composition

Solvents are described by their molecular composition (the number of each kind of atoms in the molecule), and the way in which they are arranged (structure). The phrase (the technical term is *formula*) $C_2H_3Cl_2F$ means there are two carbon atoms, three hydrogen atoms, two chlorine atoms, and one fluorine atom in a molecule of this solvent. The word *structure* simply means the way the atoms in the solvent molecule are arranged. A formula gives the atomic composition of the molecule. It does not describe the structure. The chemical name probably gives the atomic composition of the molecule. It does not describe the structure either.

A less complicated description for this solvent is HCFC. This refers to the class or family to which the solvent belongs. In this case, HCFC means hydrochlorofluorocarbon (the first “C” means chloro, the second “C” means carbon). Users of this solvent (now banned in the U.S.) know this solvent as HCFC-141b.

The arrangement of the atoms in the solvent molecule is shown in Figure 11.7 for $C_2H_3Cl_2F$. The reason chemical structures are important is that they tell users how the chemical will perform, relative to removal of certain soils whose structure is known and relative to other cleaning solvents whose structure is also known, and also relative to physical properties such as boiling point, surface tension, and flash point.

Solvent structure is relative. There are no solvent structures which are absolutely preferred except by suppliers of certain structures. Two points need to be considered when the structure of a cleaning solvent is selected.

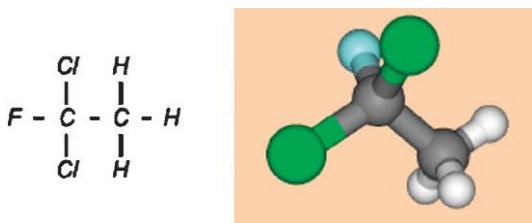


FIGURE 11.7 The structure of HCFC 141 $C_2H_3Cl_2F$ isomer b.

- Consider the structure of the solvent relative to that of the soil. Guidance is simple: match the structural features of the solvent to those of the soil (as well as they are known). Alcohols (or glycols) are good solvents for soils containing alcohol (or glycol) structures. The same is true for ester and other structures. HSP data in [Section 11.4](#) allow this to be done in an empirical manner without a detailed knowledge of chemistry. This technology is mature, having been developed and tested in the mid-1960s.
- Consider the structure of the solvent relative to the process or environment in which it will be used. Guidance here is more complicated with aspects which contradict one another. For example:
- Hydrogen and carbon atoms provide solvency and reduce inertness. They also increase the potential for ignition.
- Halogen atoms, in general, contribute the ability to suppress combustion. Fire suppressants often contain halogen atoms. Flash point values of halogenated cleaning solvents can often be unexpectedly high or non-existent.
- Fluorine atoms provide inertness and volatility and add cost.
- Chlorine and bromine atoms in molecules provide solvency for certain soils and increase concern about toxicity (especially to human kidneys or liver).
- Oxygen atoms reduce volatility (raise boiling points) through creation of hydrogen bonds among solvent molecules. Nowhere is this effect more evident than among primary alcohols (hydroxyl group attached to the end carbon atom) or glycols (hydroxyl group attached to *both* end carbon atoms). The effect is caused by inter-molecular forces between adjacent hydroxyl groups (hydrogen-bonding).
- Increased molecular weight increases boiling, and often, flash points.
- Branched or isomeric molecules have lower boiling and flash points than their relatives of the same molecular weight.
- Molecules without polar atomic components, such as hydrocarbons, are extremely poor conductors of static charge. Release of accumulated charge is necessary for safe operation or the level of static charge can increase to a point where it can produce ignition of hydrocarbon structures.
- Many solvent products are not single components. This is true for partially refined or mixed hydrocarbon solvents (e.g., hydrocarbon products whose name includes “spirits”), solvents derived from vegetable oils (soy, corn, or citrus), and solvents whose components are isomers which are difficult to separate. HFE and OS solvents are mixtures of isomers.

These and many other structures are described in [Table 11.23](#). Homologs are compounds with similar structures or functionality, but different molecular size (weight).

The solvents in this table do not have similar chemical structures. Nor do they perform solvent cleaning of the same soils to produce similar results. And the solvents included in [Table 11.23](#) are not the only ones available.

TABLE 11.23 Chemical Structures of Some Cleaning Agents

Structure	Description	Homologs
Paraffinic (straight-chain, or paraffin)	$ \begin{array}{ccccccc} H & H & H & H & H & H \\ & & & & & \\ H-C & -C & -C & -C & -C & -H \\ & & & & & \\ H & H-C-H & H & H & H & H \\ & & & & & \\ & H & & & & \\ \text{Heptane, } C_7H_{16} \end{array} $	Isopentane (C_5H_{12}), octane (C_8H_{18}) (note: there is more than one structure called isopentane)
Aliphatic (branched or isomeric)	$ \begin{array}{ccccccc} H & H & H & H & H & H \\ & & & & & \\ H-C & -C & -C & -C & -C & -C-H \\ & & & & & \\ H & H & H & H & H & H \\ \text{Hexane, } C_6H_{14} \end{array} $ Note that hexane and isopentane have the same molecular formula, but a different structure. Also, 2-ethyl hexane has the same molecular formula as octane but a different structure	2-ethyl hexane (C_8H_{18})

Aromatic	$ \begin{array}{c} H \\ \\ H - C - C - - - C - H \\ // \quad \backslash \quad \\ H - C \quad C - H \quad H \\ \backslash \quad / \\ H - C = C - H \end{array} $ Toluene, C₇H₈	NA
Alcohols	$ \begin{array}{c} H \\ \\ H \quad \quad H \\ \quad C \quad \\ H - C - - C - H \\ \quad O \quad \\ H \quad \quad H \\ H \end{array} $ Isopropanol, C₃H₈O	Isobutanol (C ₄ H ₉ O), normal butanol (C ₄ H ₉ O)
Ketones	$ \begin{array}{c} H \quad H \quad H \\ \quad \quad \\ H - C - C - C - C - H \\ \quad \quad \quad \\ H \quad H \quad O \quad H \end{array} $ Methyl ethyl ketone (MEK), C₄H₈O	Acetone (methyl methyl ketone, C ₃ H ₆ O); methyl isobutyl ketone

TABLE 11.23 Chemical Structures of Some Cleaning Agents—Cont'd

Structure	Description	Homologs
Glycol ethers	$ \begin{array}{ccccccc} H & H & H & H & H \\ & & & & \\ H-C & -C & -C-O-C & -C-H \\ & & & & \\ H & O & H & H & H \\ & & & & \\ & H & & & \end{array} $ Propylene glycol ethyl ether, $C_5H_{11}O$	Ethylene glycol ether (C_4H_9O); based on methanol, ethanol, isopropanol, or higher alcohols
Dibasic acids	$ \begin{array}{ccccccc} H & H & H & H & H & H \\ & & & & & \\ H-O & -C & -C & -C & -C & -C-O-H \\ & & & & & \\ O & H & H & H & H & O \end{array} $ Adipic acid, $C_6H_{12}O_4$	Glutaric acid ($C_5H_8O_4$), succinic acid ($C_4H_6O_4$)
Dibasic acid esters	$ \begin{array}{cccccccccccccccc} H & H & H & H & & H & H & H & H & H & H & & H & H & H & H \\ & & & & & & & & & & & & & & & \\ H-C & -C & -C & -C-O & -C & -C & -C & -C & -C & -C-O & -C & -C & -C & -C-H \\ & & & & & & & & & & & & & & & \\ H & H & H & H & & O & H & H & H & H & O & & H & H & H & H \end{array} $	Based on adipic, glutaric, or succinic acids; with methanol, ethanol, isopropanol, or higher alcohols

	Generalized dibasic acid ester, C ₁₄ H ₃₀ O ₄			
Chlorinated I	$ \begin{array}{c} Cl \quad H \\ \quad \\ Cl - C = C - H \\ \quad \\ Cl \quad Cl \end{array} $ Trichloroethylene C₂H₂Cl₄	$ \begin{array}{c} H \\ \\ H - C - H \\ \\ Cl \end{array} $ Methylene chloride CH₂Cl₂	$ \begin{array}{c} Cl \quad Cl \\ \quad \\ Cl - C = C - Cl \\ \quad \\ Cl \quad Cl \end{array} $ Perchloroethylene C₂Cl₆	NA
Chlorinated II	$ \begin{array}{c} Cl \quad H \\ \quad \\ Cl - C - C - H \\ \quad \\ Cl \quad H \end{array} $ 1,1,1 Trichloroethane C₂H₃Cl₃	$ \begin{array}{c} Cl \\ \\ H - C - Cl \\ \\ Cl \end{array} $ Chloroform CHCl₃	$ \begin{array}{c} Cl \\ \\ Cl - C - Cl \\ \\ Cl \end{array} $ Carbon Tetrachloride CCl₄	NA

Continued

TABLE 11.23 Chemical Structures of Some Cleaning Agents—Cont'd

Structure	Description	Homologs
Chlorinated III	$\begin{array}{cc} Cl & H \\ & \\ C = & C \\ & \\ H & Cl \end{array}$ Trans 1,2 Dichloroethylene C₂H₂Cl₂ Trans means that the two atoms are on the <i>opposite</i> side of the double bond. Cis means that the two atoms are on the <i>same</i> side of the double bond $\begin{array}{cc} Cl & H \\ & \\ C = & C \\ & \\ Cl & H \end{array}$ Cis 1,2 Dichloroethylene C₂H₂Cl₂	NA
Brominated	$\begin{array}{ccccc} H & H & H \\ & & \\ Br - C & - C & - C - H \\ & & \\ H & H & H \end{array}$ <i>n</i>-Propyl bromide C₃H₇Br $\begin{array}{c} H \\ \\ Cl - C - Br \\ \\ H \end{array}$ Chlorobromomethane CH₂BrCl	NA

Chloro-fluorinated	<i>Cl</i> <i>H</i> <i>F</i> <i>- C - C - H</i> <i>Cl</i> <i>H</i> HCFC 141 isomer b		NA
Chloro-fluorinated II	<i>F</i> <i>F</i> <i>H</i> <i>F - C - C - C - Cl</i> <i>F</i> <i>F</i> <i>Cl</i> Note the reversal of the fluorine (F) and chlorine (Cl) atoms on the end carbon atoms	<i>F</i> <i>F</i> <i>H</i> <i>Cl - C - C - C - F</i> <i>F</i> <i>F</i> <i>Cl</i>	NA
Fluorinated I	<i>F</i> <i>H</i> <i>H</i> <i>F</i> <i>F</i> <i>F - C - C - C - C - C - F</i> <i>F</i> <i>F</i> <i>F</i> <i>F</i> <i>F</i> C₅H₂F₁₀ HFC 43-10 isomer mee		NA

TABLE 11.23 Chemical Structures of Some Cleaning Agents—Cont'd

Structure	Description	Homologs
Fluorinated II HFE 7200	$ \begin{array}{ccccccc} F & F & F & F & & H & H \\ & & & & & & \\ F - C - C - C - C - O - C - C - H \\ & & & & & & \\ F & F & F & F & & H & H \end{array} $ C₆H₅OF₉, HFE 7200 1-ethoxy-nonafluorobutane	1-Methoxy-nonafluorobutane (HFE 7100)
Specialty <i>n</i> -MP	$ \begin{array}{c} H \\ \\ H - C - - - - C = C \\ \quad \quad / \quad \quad \backslash \\ H \quad H - C \quad C = O \\ \quad \quad \backslash \quad / \\ \quad \quad N \end{array} $ <i>n</i>-Methyl-2-pyrrolidone, C₅H₉NO	

11.5.1.2 Technical Data

Chemical structure is the key to understanding how and why solvent cleaning systems perform or do not perform as desired. Chemical structure and atomic composition determine the physical properties of cleaning solvents. This technical data is essential to users and designers of cleaning machines.

Good sources of technical, and safety/health/environmental, data include NIOSH (National Institute for Occupational Safety and Health),⁷⁴ the Internet^{75–79} and several reference books.^{80–82}

11.5.2 Cleaning/Rinsing/Drying Processes

Many functions must be completed to produce excellent cleaning work. The choice of a useful cleaning solvent was described in [Section 11.2](#) as including one that will perform solution of the soil, removal of the soil, and separation of the soil from the solvent.

But an effective cleaning process must also provide other functions. Only some are provided by the solvent. The process must augment the solvent, compensate for limitations of the solvent properties, and provide other functions which solvents cannot. For example, if the chosen cleaning solvent evaporates slowly and dry parts must be provided in a timely manner, it is the capabilities of the process equipment which must remove the liquid solvent by impingement, forced-hot gas (N_2 or air) evaporation, or some other scheme.

The limitations of solvent properties may include: surface tension being too high to allow adequate penetration of the solvent within tight space limitations; density being too low to provide adequate displacement of low-density organic soil materials and particles from within part structures; solubility limit for some soil components being too low to remove all soil components at an adequate rate; and miscibility with water being too high so that tramp moisture is not purged but accumulates within the system.

The performance valued by users is provided by the interaction of the cleaning solvent and the cleaning process.

11.5.2.1 Solvent Cleaning Processes

There are only two types of processes: those in which the solvent is vaporized, which are usually called *vapor degreasing*, and those in which the solvent is not vaporized, which are usually called *cold cleaning*. The term vapor degreasing, while commonly used, is actually a misnomer. The solvent vapor is not the degreasing agent. It is the condensed solvent vapor (liquid) which solubilizes the soil and is the degreasing agent. The virtue of this scheme (vapor degreasing) is that the parts are washed at every stage of cleaning with pristine (hopefully) soil-free distilled solvent.

Environmental regulations pertaining to emission of VOCs have become a dominant factor in the selection of one of these types of processes. This is

especially true in Europe, where VOC exempt solvents are those with high boiling points; in countries belonging to the European Union (EU), the vapor pressure at 20 °C cannot exceed 0.075 mmHg, which would not be suitable for vapor degreasing.

But within these two types of cleaning processes are a great many variations resulting from differences in design features.

11.5.2.1.1 Design Features of Solvent Cleaning Processes

Design features of solvent cleaning processes include:

- Chambers or vessels in which the solvent is contained and the parts are immersed.
- Fixtures for supporting the parts.
- Hydraulic facilities for exposing all sections of the supported parts to a pressurized flow of solvent (hot or cold).
- Means for generating pressure fluctuations so that solvent vapor bubbles are formed on the part surface. The bubbles collapse and the released energy can dislodge particles from part surfaces. This is usually done with ultrasonic or megasonic transducers.
- Means for moving (translating or rotating) the part fixtures to achieve exposure of all sections.
- Use of enhanced gravitational forces to allow cleaning solvent to drain from the parts so that dissolved soil (“drag out”) is not passed from the cleaning stages to the rinsing stages. Often the parts are vibrated to enhance separation of liquid droplets. The cleaning cycle should provide adequate time for liquid drainage, so that at least one-half of the attached cleaning solvent is removed prior to entry into the next process stage.
- A variety of independent methods to minimize exposure of the surrounding air to solvent.⁸³

Not all such features are present in every process. The features listed above bring value only when they enhance removal of soil from parts.

11.5.2.2 Rinsing Processes

Rinsing is similar to cleaning except the contaminated cleaning solvent has become the soil, and the cleaning agent is fresh solvent.

It makes no sense to implement a well thought-out cleaning operation and scrimp on planning or testing, contact time, floor space, process facilities, or quality control associated with rinsing. Too many consulting clients spare no expense when purchasing a cleaning system and literally spend nothing on facilities for rinsing. Usually, a part contaminated with soil brings no more value to end-use customers than does a part contaminated with soiled cleaning solvent.

The same features present in cleaning operations can and usually are present in rinsing operations as discussed in the previous section. In addition, design

features of solvent rinsing processes should include facilities for separating a substantial amount of whatever contaminated fluid is attached to each part before the wetted part is passed to a sequential rinsing stage or a drying stage. Separation of contaminated solvent from fresh solvent is the key performance limitation and has the most impact on cost of any step in a rinsing process.

It should be no surprise to receive a proposal from a credible supplier with a single cleaning stage and multiple rinsing stages. Standards are higher for rinsing processes than cleaning processes because they last contact the parts before drying and use.

11.5.2.3 *Drying Processes*

The unit operations involved in drying of parts have little in common with those involved in cleaning of parts. Different engineering fundamentals are involved. The unit operation of drying presents more challenge to achieve a certain level of quality than does rinsing or cleaning.

The popular definition of drying is evaporation of a liquid. There are other useful techniques for drying of parts, however, where evaporation is not the prime unit operation. Some involve removal of the liquid rinse solvent via momentum transfer where a high velocity stream of air (usually) contacts the parts and the solvent is removed as liquid droplets in a moving gas stream; use of surface-tension gradient forces between isopropanol and water to dry flat surfaces (the Marangoni drying system); use of a cosolvent such as isopropanol to remove the rinse solvent while the cosolvent is removed in a sequential process; and use of a hot inert gas such as a perfluorinated fluid to remove the rinse liquid.

An approach less likely to waste money or degrade quality is to view drying as another separation step. Here a liquid (solvent) is to be separated from a solid (the parts).

Usually drying of parts involves a necessary tradeoff. Speed and quality of drying are enhanced by use of air heated to a higher temperature. But damage to parts, and potential injury to workers by burns, can also be enhanced by air heated to higher temperature. A nominal limit is slightly above the boiling point of water.

Drying under vacuum is a method by which drying speed and quality can be enhanced without these drawbacks—except for the negative concerns associated with the cost for vacuum equipment.

11.5.2.4 *Design Features for Environmental Control*

Vapor degreasers built a generation ago were designed to efficiently and quickly contact parts with solvent. Solvents were cheap. A VOC was nearly unknown. Today, the cornerstone of a successful design of any vapor degreasing or cold cleaning unit is how solvent emissions are collected and retained

within the unit. Most cleaning solvents are smog-forming chemicals or VOCs. See Table 11.6 for a list of cleaning solvents considered by the U.S. EPA to be exempt from VOC control regulation. Not all states accept this definition. For example, perchloroethylene is exempt from VOC regulation in 49 states of the U.S., but not in California.

Earlier generations of vapor degreasers were of the open-top variety. Their use cannot be recommended for future-capitalized installations because of the reduced investment and improved maintenance records associated with enclosed cleaning machines which essentially avoid emissions of VOC cleaning solvents.

Consequently, contained cleaning is of significant and growing interest. This means that the cleaning, rinsing, and drying steps are individually completed under a vacuum chamber.⁸³ Leaks of VOC material are contained within the cleaning machine, not released to the environment. Contained may also mean that the entire cleaning, rinsing, and drying process is enclosed in an “envelope” which is a pressurized system. Potential leaks of VOC material are recognized as a loss of containment pressure. Detection of that event mandates human inspection or enables automatic shutdown.

11.5.2.5 Multiple-stage Processes

Equilibrium is an important concept in thermodynamics. When two systems are in equilibrium, there is no net transfer between them. This occurs when the systems are at the same condition.

In other words, to transfer soil from parts to a solvent, the level of soil in the solvent must be less than the equilibrium solubility limit. When the soil level in the solvent is near the equilibrium solubility limit, only fresh solvent can allow removal of more soil by solution. This occurs in a multiple contact stage of solvent with parts.

The flow diagram of a typical sequential multiple-stage cleaning system is shown in Figure 11.8. Note how five cleaning stages arranged in sequence produce no improvement in cleanliness between the second and the fifth stage.

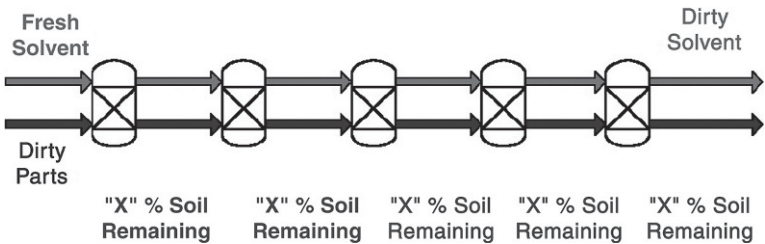


FIGURE 11.8 Multiple-stage cleaning system with “dirty” solvent contacting soil-free parts in the last stage.

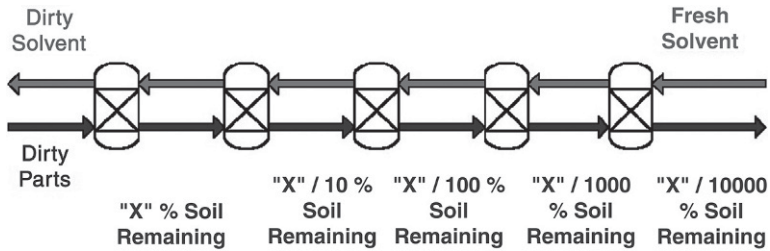


FIGURE 11.9 Multiple-stage cleaning system with fresh clean solvent contacting soil-free parts in the last stage.

Stated still another way, if soil-free parts are rinsed with solvent containing dilute amounts of soil, the parts will certainly become no cleaner and will likely become more soiled.

A cardinal principle of solvent cleaning (as well as aqueous cleaning) is that in order to produce pristine parts, the last contact stage (the final point of contact of the parts with process) must be free of whatever is being removed from the parts. *Pristine* in cleaning work means free of everything: a pristine part has no soil residue and all the cleaning agent (solvent or water-based) has been removed. This is shown in the flow diagram in [Figure 11.9](#), where the last solvent contacting the parts is pristine (fresh).

Note how the level of soil remaining on the parts is diminished by an artificial factor of 10 in each of the five stages within this cleaning system. While this is probably unrealistic in practical equipment, the improvement in part cleanliness can certainly be seen. In theory, thermodynamics teaches that total soil removal takes an infinite number of contact stages.

11.5.2.6 Selection of Design Features

Rules for the selection of types and features of solvent cleaning processes have been developed by the author based on years of practical experience. They include:

1. A given project for cleaning can be accomplished with many different choices of cleaning process and solvent. There is no single approach that is required for any cleaning project.⁸⁴
2. Almost any project for cleaning can be done with any combination of process and solvent, but the consequences may be severe. Too many times users have made choices which were commercially attractive but not technically sound.
3. These users suffered because they accepted high costs of materials or equipment, or operating labor; performance to a standard of quality which was only acceptable and not consistently achieved; vulnerability to current

- and future environmental/health/safety regulations; and purchase of equipment which did not produce cleaned parts at the required production rate.
4. Base the choice of type and design features of solvent cleaning process on the characteristics of the parts.
 5. Choose the rinsing process according to the level of quality needed for the finished parts. For example, the parts will be no freer of soil components than is the soil concentration in the rinse fluid with which they are last contacted. Often the rinsing process can have more steps than the cleaning process, add more investment in process equipment, require more labor and floor space, and be of greater concern.
 6. Choose the drying process based on what will be done next with the parts. A secondary consideration is the level of quality needed for the finished parts. For example, in the drying process, the rinse fluid (no matter how pure) is a soil.
 7. Choose the number of cleaning, rinsing, and drying stages based on how clean they have to be. The number of each may, and probably will, be different. As above, the cleaning solvent should be chosen to match the HSP, as nearly as possible, of the soils on the parts.
 8. Combine your choices and screen them based on their environmental and cost consequences. This is easier said than done, but is specific to every application. Many users write simple spreadsheets which allow good decisions. The services of a consultant may be of value here.
 9. Finally, select the supplier to implement these choices based on their perceived ability to meet your needs. For example: do not use prices as screening criteria. Low price when the system is not working is not a need; returning the system to quality service is the prime need. Do not waste your time obtaining financial quotations from every supplier—spend your time learning about how they have (or have not) met the needs of others, and can (or cannot) meet your needs.

11.6 CONTROL OF QUALITY IN SOLVENT CLEANING

Many users consider solvent cleaning to be more forgiving of mis-operation than aqueous cleaning. This author's experience is consistent with that belief. Yet solvent cleaning is much more than applying a solvent to a part, removing the solution of soil and solvent, and drying the part.

Adherence to certain metrics and standards is necessary to achieve, and continually achieve, the desired quality of soil removal. These metrics and standards are derived from study of the three fundamental unit operations being performed in solvent cleaning. They are shown in [Table 11.24](#).

In addition to management of quality cleaning through attention to standards and metrics, it is also necessary to pay attention to examination of the cleaned parts. Inspection testing, covered elsewhere in the literature, is not the only necessary inspection activity. Others include the following:

TABLE 11.24 Standards and Metrics for Solvent Cleaning

Unit Operation ¹	Standard	Methods	Metric(s)
Solution ² of the soil ³ in a solvent	The environment producing the solutioning function is constant	Maintain bath temperature at the boiling point via adequate input of heat	In process measurement and set-point control
		Maintain constant soil level in bath via removing soiled solvent at a rate equivalent to the input rate	1. Measured purge rate vs. set-point; 2. Measured change vs. set-point in condensate level in distillation system over time 3. Measured dynamic surface tension⁴ vs. set-point. Dynamic surface tension is a strong function of impurity (soil) level in liquids 4. Maintain stabilizer level within limits recommended by solvent supplier, using technology provided by that supplier
		Maintain constant flow rate of solvent pumped through a nozzle on to all areas of the parts	1. Measured pressure drop vs. set-point across the injection nozzle 2. Measured pressure drop vs. maximum across the filter in the flow line
Augmentation of the solutioning process by applying mechanical force to dislodge particles, force solvent into volumes with low clearances, or intermix solvents with polymeric residues so as to increase the degree of swelling by solvent ⁵	The same level of mechanical force is applied to every area of every part every cleaning cycle	Maintain alignment of nozzle(s) with surfaces of parts	1. Validated when cleaning machine is first put into service, and verified periodically when machine is out of use 2. Use of baskets with every cleaning cycle in which parts are arranged to allow appropriate contact with fluid jets

Continued

TABLE 11.24 Standards and Metrics for Solvent Cleaning—Cont'd

Unit Operation	Standard	Methods	Metric(s)
Heating of the cleaning process materials so as to improve performance ⁶	In vapor degreasing, heat is only being applied through maintenance of the cleaning bath at the solvent boiling point	In no case should parts ever be heated prior to entry into the cleaning chamber ⁷	Maintain cycle time in the range of seconds, unless liquid immersion is part of the cleaning process
	In cold cleaning, parts may be heated prior to entry into the machine to enhance solubility	Heat parts prior to entry into the machine to enhance solubility	Monitor cleaning bath temperature vs. set-point to assure that heat input rate is at least matched by heat removal rate so that bath temperature does not approach boiling point

¹Aqueous cleaning technology involves the same three unit operations. In aqueous cleaning, mechanical force plays a much more dominant role. In solvent cleaning, solutioning plays the dominant role.

²Solutioning is not a process in which an unlimited amount of soil can be dissolved within the volume of solvent contained in the stage(s) of the cleaning machine. As with solution of salt in water, there is a solubility limit for every soil (or combination of soil components) in any solvent. In general, this solubility limit will not be known without data.

³Soil removal is a process which includes a diffusion-based mechanism. If the composition of the cleaning bath is near (or at) the solubility limit, the rate of mass transfer by diffusion from the part surface to the bulk solvent-soil mixture in the bath will approach (or equal) zero. In simple terms, dirty parts cannot be cleaned with dirty (soil-rich) solvent. In addition, soil-rich solvent may deposit soil previously removed from one part area to another part area. This is the opposite of cleaning!

⁴See Ref. 85.

⁵Often cured polymeric residues are resistant to solutioning by solvents, even though this might be expected from a study of solubility parameters (see Section 11.4). A polymer swollen with solvent is usually less intractable to mechanical force or partial solutioning than is one not organized as a network of polymeric molecules in a matrix of solvent.

⁶Heating may or may not be included in cold cleaning operations. The normal definition of cold cleaning includes the provision that the solvent may be heated to less than its boiling point.

⁷Conventional vapor degreasing, where parts are not immersed in liquid solvent, is a process in which nearly pure solvent is produced by condensation of vapor on part surfaces when the parts are colder than the boiling point of the solvent. Such condensation ceases when the parts have been heated by the hot vapor to the solvent boiling point. For metal parts, such heat up normally requires only seconds to no more than a minute. Beyond that time, additional cleaning may occur, but it will not be with pristine freshly-condensed solvent.

1. Process equipment must be as clean prior to use as the cleaned parts are required to be after processing. Completion of several processing cycles without parts can be very useful after the equipment is inspected for cleanliness.
2. Designers of cleaning machines must provide the capability to remove soil from parts *and* to remove soil from the machine. Inspection is necessary to ensure that soil is not being removed from parts but not from the machine.
3. Similarly, the processing environment and packaging must also be at least as free of suspended particles and biological matter as are the cleaned parts. It is pointless to produce excellent cleaning work and spoil it via either post-processing contamination or less than pristine packaging.
4. Upstream activities must also be monitored. If the parts are soiled with a new soil, the chosen cleaning solvent probably will not perform as expected. This often happens when cost-reduction activities are of paramount interest.
5. Finally, it is not enough to monitor existing performance. Current performance must be placed within the perspective of past performance. The metrics noted in [Table 11.24](#) should be compared to past values. Here some methods of statistical process control, such as CUSUM, can indicate the onset of off-quality operation before its consequence becomes severe.⁸⁵

11.7 AVOIDING COMMON MISTAKES

Solvent cleaning is more than applying a solvent to a part; it is a process. Information in [Section 11.5](#) briefly addresses design features of the three unit operations of which solvent cleaning processes are comprised: cleaning, rinsing, and drying.

Awareness of and fidelity to the guidelines within that section are not sufficient, however, to avoid critical mistakes in management of solvent cleaning applications. Users can learn from, and copy, the experiences of other users.

Some of the issues producing these mistakes are not technical, but are psychological.

Mistake 1: Unwarranted repetition. The most significant, and most common, mistake users make is to repeat what has been done by others, even though the factors in the user's application are not the same as the factors in the applications of others. Certainly there is psychological comfort in relying on what has been successful. But dissatisfaction is likely if the basis for an application is not understanding the key factors which affect performance in a supposedly successful example.

Mistake 2: Supplier selection. There is only one good reason for selection of suppliers of cleaning equipment and solvents. This is: the chosen supplier very much wants your application to be successful so that it will bring them additional business.

Make identification of this firm of primary concern when interviewing suppliers and reviewing their proposed equipment, solvent offerings, and service. Too many unsatisfied clients have purchased what they considered to be the proper equipment, only to find that the supplier of that equipment was chiefly concerned about the profit from their sale and significantly less concerned about the success of the application.

Today, few firms can propose a unique offering. Practical and technical experience, desire for your success, and follow-through are much more difficult for competitive suppliers to match.

Mistake 3: Focus on price. No user wants to pay a price higher than that offered by the competition. In fact, firms commonly require their employees to always choose the firm offering the lowest price. Why would a user not seek that price? The reason is the majority of the true cost of cleaning articles is not just the purchase price of the equipment or the solvent. It is the cost of poor performance after purchase. Consequently, users should seek the offering of equipment, service, and solvent which appears to offer the greatest chance of success. A successful operation will ultimately produce the lowest financial burden because products are nearly always significantly worth more than the cost of cleaning them. Users should be free to accept this approach because pricing in the cleaning industries is quite competitive. With the exception of large machines used to clean large volumes of metal parts, no single firm dominates a market.

Mistake 4: Absence of attention to waste disposal. The cost of disposal of waste solvent (solvent and soil) is typically higher than the cost of purchase of fresh solvent by a factor of from 10 to 100. This is especially true if the waste solvent is classified as hazardous because of low flash point ($<140^{\circ}\text{F}$) or of the contained soil. Expensive “designer” solvents, HFCs, HFEs, and some HCFCs, should always be recycled after purification by removal of soil. They should not be discarded as waste because of the high purchase cost. Reputable distributors will offer this service.

Mistake 5: Ignorance about costs of environmental compliance. This mistake is often committed as a corollary to mistake 4. A cleaning solvent should never be selected based solely on price. Costs of disposal and environmental compliance usually dwarf the cost of solvent purchase. Costs of environmental compliance can range from the legal cost of obtaining a permit to the excessive cost of abandoning an operation which is not sustainable because of failure to comply with environmental regulations.

The above mistakes are offered as negative examples because space does not permit publication of application details. The above can be summarized in a single point of view. Most firms—other than those who provide the service of contract cleaning—are not in business to complete the work of cleaning, but to produce and sell articles at a profit. Cleaning is a means to that end. The focus of management of cleaning work should be for it to be so successful that it is essentially ignored. That is the true position of lowest cost.

11.8 THE FUTURE OF SOLVENTS AND CLEANING

The future is always driven by forces. Sometimes those forces are recognizable. Sometimes they are what could have or should have been recognized. In this section, the focus will be on the following forces, which may materially affect the future of solvents and solvent cleaning: customer preferences, environmental regulations, and innovation.

11.8.1 Customer Preferences

Choices of cleaning solvents described here may appear to some users as an infinite and confusing matrix. There are at least 200 primary solvents, not counting blends and azeotropes formulated from some of those solvents. Users in the U.S. favor VOC-exempt solvents that offer low reactivity with UV light and low boiling points so they can be used in vapor degreasing applications. In contrast, users in Europe favor VOC-exempt solvents with high boiling points, and consequently cannot be used in vapor degreasing applications. But these choices are not enough.

There is no cleaning solvent about which it can be said that there is no major drawback in its use. (Similarly, this author believes there is no aqueous cleaning agent about which the above statement cannot be legitimately made.)

Some of the unfulfilled goals of users of cleaning solvents are solvents with, or which are,

- VOC exempt, in the region, country, or area of use.
- Fewer concerns about health, safety, and environmental impact.
- No flash point, or ideally, no flammable limits.
- Solvency which matches more commercial soils. Technically, this means HSP values are more closely aligned with those of soils.
- Lower surface tension so that penetration of complicated structures can be made more repeatable and reliable.
- Higher density so that soils of lower density (oils, greases, etc.) can be more completely flushed.

These preferences are not likely to be achieved quickly, either practically or in a financially sound way.

Basically, the market for cleaning solvents is not likely to support commercial product development; medical, toxicological, and environmental testing; and application development of significant new generations of cleaning solvents. The reason is that the financial stake for solvent producers is not sufficient to justify the expense for that development. Stated another way, the market for cleaning solvents is too small to support future development of new solvents.

The solution to this commercial dilemma is blends and azeotropes. The latter are blends of solvents which have a single boiling point. Control of vapor degreasing operations becomes simpler. This means formulation of new

products. The feedstocks are solvent products which have passed medical, toxicological, and environmental screening hurdles and whose cost of commercial manufacture is financially justified by existing business volume. The new products are those which may generate less regulatory concern, have the desired HSP values, have lower surface tension and higher density, and which may not be flammable.

This trend is recognizable since at least the last millennium. Examples are:

- Azeotropes of alcohols or esters with HFCs and HFEs to produce solvents for vapor degreasing which have the desired HSP values.
- Blends of flammable solvents, such as trans-1-2-dichloroethylene, acetone, methyl acetate, isopropanol, and D-Limonene, with HFEs and HFCs which are not flammable so that the commercial product passes the flash point screening test.
- Blends and azeotropes of VOC-exempt solvents, such as acetone, methyl acetate, HFEs, or HFCs to produce solvents for vapor degreasing which are not flammable.
- Blends of naturally-produced solvents, such as soy-based products, with other solvents, to produce commercial products whose HSP values have more broad commercial use.

U.S. patent coverage is claimed for many of these blends and azeotropes.

The only brake on this imperative to offer cleaning solvent blends which comply with customer preferences is the decline of the overall desire to do cleaning, or do cleaning with solvents.

11.8.2 Environmental Regulations

The direction of change is certain. It is the path and pace of change which is less so.

With only rare exceptions, the direction of change in environmental, safety, and health regulations which affect solvent cleaning is toward less: that is, less emissions of chemicals (solvents) to the air, water, or landfills; less use of chemicals; less contact of chemicals by humans; fewer grants of permits by local regulatory authorities to conduct solvent cleaning operations; and reduced opportunities by choice to do solvent cleaning.

As an example of the rarely-seen exemption, the U.S. EPA increased its exposure limit for use with HCFC 225 ca/cb from 25 to 50 ppm based on later-developed toxicity information.⁸⁶

Does this mean that solvent cleaning is being outmoded, applications are declining, and future cleaning will be done with “clean” beams of energy? Perhaps this is so, if the time horizon is measured in generations. But if the time horizon is measured in years or decades, this author believes that innovation will continue to provide protection of the environment and workers while enabling the cleaning of parts with solvents.

11.8.3 Innovation

11.8.3.1 Business Innovation

Until recently, innovation produced enclosed machines for solvent cleaning based on either containment by vacuum or by seal against pressure.

Today, innovation is reducing the investment associated with such machines. Two suppliers have developed *basic* vacuum-based machines that will sell in the U.S. for less than U.S. \$50,000. To be sure, this machine is not full-featured or suitable for all applications. But its appearance represents business innovation because previously-offered products had a price twice as high.

By the end of this decade in which this book is published, this author believes that the new investment required for vacuum machines will be the same or less than that required for open-top machines which meet local environmental requirements.

Stated another way, despite significant commercial failure and reorganization, vacuum-based solvent cleaning machines will displace workhorse open-top machines within two decades after the former was patented.⁸⁷

11.8.3.2 Technical Innovation

Future cleaning solvents might be non-organic ionic liquids which do not evaporate and produce pollution. Ionic liquids are organic salts with melting points under 100 degrees (either °C or °F), often even lower than room temperature.

Recently they have been employed more and more as substitutes for the traditional organic solvents in chemical reactions—though not yet for solvent cleaning.^{88–92}

Most of the familiar cleaning solvents (e.g., water, ethanol, hexane) are molecular. That is, regardless of whether they are polar or non-polar, they are basically constituted of molecules. Ionic liquids are different. They are composed of ions, not molecules. Unlike molecular liquids, regardless of the degree of association, ionic solvents are basically constituted of ions.

Ionic liquids may allow some unique advantages in cleaning work because of their structure. But they will certainly add drawbacks. They are not volatile. They do not evaporate. All ionic solvents would, supposedly, be VOC exempt in both the U.S. (unreactive with UV light) and in Europe (very high boiling). However, drying of parts would become more challenging.

11.8.3.3 Regulatory Innovation

Innovation in modeling the atmospheric reactions which lead to formation of smog would strongly affect those doing solvent cleaning work.

Solvent cleaning shares a situation with a host of other applications such as the baking of bread, driving of automobiles, painting of surfaces, and storage of

fuels. All these applications emit VOCs which react in the troposphere to produce atmospheric smog.

Smog is formed by a complex and not completely understood reaction scheme (see Section 11.2). At least four factors dominate minimization of smog formation. They are: release of reactive (with UV light) VOC chemicals; release of oxides of nitrogen from combustion processes; reaction of the two and other species; and air currents in the troposphere which control the concentration of reactants and products.

U.S. EPA maintains a strong interest in modeling the interaction of emissions of chemicals with air currents—the precursors of smog.⁹³ Environmental regulatory agencies in other countries and continents share that interest.⁹⁴ The state of this work does not yet support specific and unique regulation in the U.S.

Today, few users substitute cleaning solvents to achieve exemption from regulations affecting VOC management. Today, users in the U.S. have access to only a few VOC-exempt cleaning solvents. Most solvents are not so exempt.

An innovative regulatory scheme which classified solvents by their propensity to form smog would increase choices for those doing cleaning work, and reduce net emissions of smog-forming chemicals. For example, under the relative scale proposed by Carter and co-workers,⁹⁵ users could choose cleaning solvents based on both expected cleaning performance from HSP information and a ranking of potential to form smog accepted by U.S. regulatory agencies.

As of this writing, it is not yet clear if such a scheme would attract the necessary bureaucratic and political support in the United States. But, the technology to do so is available to users and there is no promulgation *against* doing so.

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